VIGNAN INSTITUTE OF TECHNOLOGY AND SCIENCE
Near Ramoji Film City, Deshmukhi Village, Pochampally Mandal, Yadadri Bhuvanagiri Dist.
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AN AUTONOMOUS INSTITUTION

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CrossTongueGuard: Detecting Cyberbullying in Code-Mixed Conversations

A MAJOR PROJECT STAGE-II REPORT

Submitted by

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Under the guidance of

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Assistant Professor

in partial fulfillment of the degree of

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE AND ENGINEERING (AI & ML)

April 2026

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CERTIFICATE

This is to certify that the project work titled **"CrossTongueGuard: Detecting Cyberbullying in Code-Mixed Conversations"** submitted by **Pathlavath Bhaskar (22891A6643)** in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering (AI & ML) from Vignan Institute of Technology and Science, Deshmukhi, is a record of bonafide work carried out by him under my guidance and supervision.

The results embodied in this project report have not been submitted in any university for the award of any degree, and the results are achieved satisfactorily.

Mr. K.Sudhakar
Project Guide

Dr. J. R. V. Jeny
Head of the Department

External Examiner

DECLARATION

I hereby declare that the project entitled **CrossTongueGuard: Detecting Cyberbullying in Code-Mixed Conversations** is bonafide work duly completed by me. It does not contain any part of the project submitted by any other candidates to this or any other institute of the university. All such materials that have been obtained from other sources have been duly acknowledged.

Pathlavath Bhaskar
(Reg.No. 22891A6643)

ACKNOWLEDGEMENT

The fulfillment that accompanies the successful completion of my project, this is a moment to convey my heartfelt gratitude to all those who have played an instrumental role in my major project's stage-II's realization. Sincere thanks are extended to **Dr. L. Rathaiah**, the esteemed Chairman of Vignan Group of Institutions.

My expression of gratitude further extends to **Mr. Boyapati Shravan**, the distinguished CEO of Vignan Institute of Technology and Science, Hyderabad.

I also extend my heartfelt gratitude to **Dr. G. Durga Sukumar**, our esteemed Principal, for generously providing me with the necessary facilities and resources that were instrumental in the successful undertaking and completion of this major project.

My profound gratitude goes to **Dr. J. R. V. Jeny**, Professor and Head of the Department, whose unwavering encouragement and support have been a source of inspiration throughout the course of my project. Her guidance and mentorship have played a pivotal role in shaping my project and fostering a conducive learning environment. Her constant supervision, and necessary information regarding the major-project, as well as her support in completing it. I am deeply indebted to her for her invaluable contributions and for being a guiding light on my academic journey.

I am also very grateful to the project coordinator **Dr. S. Godwin**, Associate Professor, for his help in solving problems during my major project stage-II.

My sincere thanks to my project guide **Mr. K. Sudhakar**, Assistant Professor,for his guidance,insightful feedback, and continuous support throughout the duration of the major project stage-II, as well as his support in completing it.

Last but not least, I wish to express my gratitude and thanks to friends and beloved parents for their support and help.

ABSTRACT

Cyberbullying has emerged as a recent concern in the global online communication platform, causing emotional distress and psychological trauma to individuals. The detection of cyberbullying is particularly challenging in the multilingual code-mixed online social media communication platform, where individuals communicate in multiple languages in a single message and use informal language, transliteration, and regional slang. These language complexities reduce the efficiency of traditional text classification techniques. To address this challenge, this paper proposes CrossTongueGuard, a systematic framework for the detection of cyberbullying in multilingual code-mixed communication platforms. The framework employs systematic preprocessing techniques, semantic context-based annotation, and multi-aspect analysis based on intent, harm, and aggression.The framework integrates traditional machine learning models and multilingual transformer models to efficiently extract both statistical and contextual features. The primary aim of this research is to develop a scalable and language-agnostic cyberbullying detection framework that can handle real-world multilingual social media data efficiently and establish a safe online communication environment.

Keywords: Code-Mixed, Semantic annotation, Multilingual Transformer Models, Contextual Features,Multiple Languages.

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List of Abbreviations

AGD	Aggression Detection
AI	Artificial Intelligence
AML	Multi-Aspect Learning
BERT	Bidirectional Encoder Representations from Transformers
CM	Code-Mixing
CMLP	Code-Mixed Language Processing
CTG	CrossTongueGuard
DL	Deep Learning
IDH	Intent to Harm
mDeBERTa	Multilingual DeBERTa
MuRIL	Multilingual Representations for Indian Languages
NLP	Natural Language Processing
RPD	Repetition Detection
SN	Script Normalization
TRN	Transliteration
XLM-R	XLM-RoBERTa

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Chapter 1

INTRODUCTION

The rapid growth of social media platforms has significantly transformed the way people communicate and share information online. Platforms such as Twitter, Facebook, Instagram, and YouTube allow users to express opinions, interact with others, and participate in discussions. However, the widespread use of these platforms has also led to the rise of harmful behaviors such as cyberbullying, where individuals use digital communication to harass, threaten, or insult others[16],[12]. Cyberbullying can have serious psychological and emotional consequences for victims, including anxiety, depression, and reduced self-esteem.

With the rapid growth of online communities, the amount of user-generated textual content has increased significantly. Millions of comments, posts, and messages are created every day across different platforms[11],[15]. While this growth has improved communication and knowledge sharing, it has also created challenges related to content moderation and online safety[9].

Along with the benefits of social media, there has also been a significant rise in negative behaviors such as cyberbullying. Cyberbullying refers to the use of digital platforms to harass, insult, threaten, or target individuals through harmful messages or comments. Victims of cyberbullying often experience emotional distress, anxiety, depression, and other psychological impacts.In this project, a cyberbullying detection system is developed using both traditional machine learning algorithms and transformer-based deep learning models[14]. The machine learning models used include Logistic Regression, Naïve Bayes, and Support Vector Machine (SVM). In addition, advanced transformer models such as MuRIL, XLM-RoBERTa, and mDeBERTa are used to capture contextual semantic information from textual data[2],[4]. The dataset is first preprocessed and analyzed through exploratory data analysis, followed by dataset balancing and semantic context-based annotation. Feature extraction techniques are then applied, and the models are trained to classify text into cyberbullying or non-cyberbullying categories.

Detecting cyberbullying manually is challenging due to the massive volume of content generated every second on social media platforms. Traditional moderation techniques rely heavily on human moderators, which is time-consuming and inefficient for large-scale online environments[7]. Therefore, automated systems that can accurately detect and classify cyberbullying content are essential to ensure safer online communities. Machine learning and natural language processing (NLP) techniques provide effective solutions for analyzing textual data and identifying harmful or abusive language[8].

Code-mixed Hindi–English (commonly called Hinglish) introduces complexities such as inconsistent grammar, transliteration of Hindi words into Roman script, phonetic spellings, slang usage, and region-specific vocabulary. For example, the word "pagal," may appear as "paagal," "pgl," or "pagl," while abusive expressions often have countless spelling variations[1]. Such patterns make cyberbullying detection far more challenging than in monolingual English datasets. As a result, there is a growing need for robust, intelligent systems capable of understanding, interpreting, and detecting harmful behavior in code-mixed environments.

Code-mixing refers to the practice of blending two or more languages within a single sentence or conversation. This form of expression is highly prevalent among Indian social media users, who often combine English words, Romanized Hindi terms, emojis, slang, and phonetic spellings to create informal and dynamic communication[13],[1].Traditionally, online platforms rely on human moderators to monitor and remove harmful content. However, due to the enormous amount of user-generated data produced every second, manual moderation is not sufficient to handle the growing volume of comments. This limitation highlights the need for automated systems that can quickly detect and classify abusive content.

Artificial Intelligence (AI) and Natural Language Processing (NLP) techniques provide effective solutions for analyzing large volumes of textual data. These technologies enable computers to understand human language, identify patterns, and classify text based on its meaning. By using AI-based models, it is possible to automatically detect cyberbullying comments in online platforms[10].

The effectiveness of a cyberbullying detection system largely depends on the quality of the dataset used for training machine learning models. A well-prepared dataset that contains clean textual data and meaningful labels allows models to learn patterns associated with abusive or harmful language. Therefore, proper dataset construction, preprocessing, and annotation are essential steps in building a reliable detection system.The effectiveness of a cyberbullying detection system largely depends on the quality of the dataset used for training machine learning models. A well-prepared dataset that contains clean textual data and meaningful labels allows models to learn patterns associated with abusive or harmful language. Therefore, proper dataset construction, preprocessing, and annotation are essential steps in building a reliable detection system.

Exploratory Data Analysis (EDA) is an important step in understanding the characteristics of the dataset before training machine learning models. Some techniques such as correlation analysis, distribution plots, and box plots, researchers can analyze patterns present in the dataset. These visualizations help in identifying relationships between features, detecting outliers, and understanding the distribution of text-based features such as comment length.Another important factor in classification tasks is dataset balance. If the dataset contains significantly more examples of one class compared to another, the trained model may become biased toward the majority class. To overcome this problem, balancing techniques are applied to ensure that both cyberbullying and non-cyberbullying classes are fairly represented in the dataset.

Machine learning algorithms play an important role in detecting cyberbullying from textual data. Traditional algorithms such as Logistic Regression, Naïve Bayes, and Support Vector Machine (SVM) can learn patterns from labeled text data and classify comments into cyberbullying or non-cyberbullying categories.In addition to traditional machine learning techniques, transformer-based deep learning models have shown remarkable performance in natural language processing tasks[3]. Models such as MuRIL, XLM-RoBERTa, and mDeBERTa are capable of capturing contextual relationships between words and understanding the semantic meaning of text[5],[6].

To enhance the dataset quality, semantic context-based annotation is used in this project. The textual data is labeled based on different semantic attributes such as intent, harm, and aggression. Based on these attributes, the final label indicating whether the comment represents cyberbullying or not is assigned.To evaluate the performance of the proposed system, multiple machine learning and transformer models are trained and tested on the dataset. The models are evaluated using standard classification metrics such as accuracy, precision, recall, and F1-score to measure their effectiveness in detecting cyberbullying comments.

The main contribution of this work lies in the use of semantic context-based annotation combined with multiple machine learning and transformer models for cyberbullying detection. By incorporating additional semantic labels such as intent, harm, and aggression, the dataset provides richer contextual information that improves the classification process.

Overall, this project focuses on developing an intelligent cyberbullying detection framework that leverages both traditional machine learning algorithms and advanced transformer-based models. Through proper dataset preparation, semantic annotation, and comprehensive model evaluation, the proposed approach aims to improve the accuracy of cyberbullying detection and contribute to creating safer online communication environments.

1.1 Purpose of the Project

The primary purpose of this project is to develop an automated system capable of detecting cyberbullying in online comments using machine learning and transformer-based natural language processing techniques. Social media platforms have become an important medium for communication, where millions of users share their opinions, experiences, and ideas through comments and posts. However, the open nature of these platforms has also led to the rise of harmful behaviors such as cyberbullying. Detecting such harmful content manually is difficult due to the large volume of online interactions. Therefore, this project aims to design an intelligent system that can automatically analyze textual content and identify cyberbullying comments.

Another purpose of this project is to study and compare the performance of different classification algorithms for cyberbullying detection. The project evaluates traditional machine learning models such as Logistic Regression, Naïve Bayes, and Support Vector Machine, along with advanced transformer-based models including MuRIL, XLM-RoBERTa, and mDeBERTa. By implementing and evaluating multiple algorithms, the project aims to determine which models are most effective for detecting harmful language in social media text.

The project also focuses on improving the dataset quality through proper preprocessing and semantic annotation techniques. Raw text data collected from online platforms often contains noise such as special characters, repeated words, emojis, or irrelevant symbols. These elements can affect the performance of classification models. Therefore, data preprocessing techniques such as cleaning, normalization, and tokenization are applied to prepare the dataset for further analysis. Additionally, semantic context-based annotation is used to assign meaningful labels to the dataset.

Another important purpose of this project is to analyze the dataset using exploratory data analysis techniques. By examining the distribution of comments, correlations between variables, and text characteristics such as comment length, it becomes easier to understand the nature of the data. Visualization techniques such as box plots, probability density functions, and correlation heatmaps help in identifying patterns and relationships within the dataset. This analysis helps researchers gain better insights into the dataset before training machine learning models.

The project also aims to address the issue of dataset imbalance, which is common in cyberbullying detection tasks. In many datasets, the number of non-cyberbullying comments is significantly higher than cyberbullying comments. If this imbalance is not handled properly, machine learning models may become biased toward the majority class. Therefore, dataset balancing techniques are applied to ensure that the models learn meaningful patterns from both classes.Another key purpose of this project is to evaluate the performance of the trained models using standard classification metrics. Metrics such as accuracy, precision, recall, and F1-score are used to measure the effectiveness of the models in identifying cyberbullying comments. These evaluation metrics provide insights into how well the models perform in distinguishing harmful comments from normal comments.

Furthermore, the project aims to demonstrate the practical implementation of cyberbullying detection through a user-friendly application. A simple web interface is developed using Streamlit, allowing users to input comments and receive real-time predictions about whether the comment contains cyberbullying content. This application shows how machine learning models can be integrated into real-world systems for monitoring and filtering harmful online comments.

Finally, the overall purpose of this project is to contribute to safer digital communication environments by developing an intelligent cyberbullying detection framework. By combining data preprocessing, semantic annotation, machine learning models, and transformer-based architectures, the project demonstrates how artificial intelligence can be used to address social challenges related to online harassment and harmful communication.

1.2 Scope of the Project

The scope of this project focuses on developing an automated cyberbullying detection system that analyzes textual comments from online platforms and classifies them as cyberbullying or non-cyberbullying. With the increasing use of social media platforms for communication and interaction, there is a growing need for intelligent systems that can monitor and filter harmful content. This project aims to address this need by implementing machine learning and transformer-based models capable of identifying abusive or offensive language in user-generated text.

The project involves collecting and preparing a dataset of online comments and performing various preprocessing techniques to clean and normalize the textual data. The scope includes applying exploratory data analysis to understand the characteristics of the dataset and identify patterns within the comments. Additionally, semantic context-based annotation is used to enrich the dataset by incorporating attributes such as intent, harm, and aggression, which help improve the quality of the classification process.

Another important aspect within the scope of this project is the implementation and comparison of multiple classification algorithms.This project primarily covers text-based cyberbullying detection using both traditional machine learning techniques and advanced transformer-based models. It includes data preprocessing, feature extraction using TF-IDF, and classification. Traditional machine learning algorithms such as Logistic Regression, Naïve Bayes, and Support Vector Machine are used along with advanced transformer-based models including MuRIL, XLM-RoBERTa, and mDeBERTa. These models are trained and evaluated to determine their effectiveness in detecting cyberbullying in textual data.The project also includes the evaluation of model performance using standard classification metrics such as accuracy, precision, recall, and F1-score. Confusion matrices and ROC curves are analyzed to better understand the performance of each model and identify the best-performing approach for cyberbullying detection.

Furthermore, the scope of this project extends to the development of a simple user interface using Streamlit. This interface allows users to input text comments and receive predictions from the trained models. The application demonstrates how the cyberbullying detection system can be integrated into real-world platforms to automatically analyze comments and block harmful messages before they are displayed.Although the project focuses primarily on textual data, the proposed system can potentially be extended in the future to support multimedia content such as images, videos, or audio-based harassment detection. Additionally, the system can be integrated into social media platforms, online discussion forums, and comment moderation tools to improve digital safety and reduce the spread of abusive content.

Overall, the scope of this project is limited to developing and evaluating an automated cyberbullying detection framework using machine learning and transformer models, while demonstrating its practical implementation through a prototype application.

1.3 Applications

The proposed CrossTongueGuard framework has a wide range of practical applications across multiple domains where multilingual and code-mixed communication is prevalent. Its ability to detect aggression, repetition, and intent to harm in complex linguistic environments makes it particularly valuable for modern digital platforms and socially impactful initiatives.The model's ability to understand code-mixed language also benefits organizations working with multilingual user bases, helping maintain healthy online communities.

One of the primary areas of application is in Social Media Content Moderation. These system can be integrated into platforms such as Facebook, Instagram, Twitter, and YouTube to automatically identify harmful or abusive code-mixed comments. By providing early detection, it supports moderators in filtering toxic content and maintaining safer online communities.

Messaging and Chat Applications are another ideal application for this system. Popular communication tools like WhatsApp, Telegram, Snapchat, and Discord often contain mixed-language conversations. CrossTongueGuard can monitor group chats, flag harmful messages, and provide alerts in environments where automated moderation is currently limited.

In Educational Institutions, such as Schools, colleges, and universities can deploy the system to monitor cyberbullying among students on academic portals, discussion forums, or institutional social groups. This helps educators identify patterns of harassment and provide timely interventions to prevent emotional harm.

In Mental Health Support Systems Cyberbullying often contributes to anxiety, depression, and psychological stress. Mental health organizations can use the system to track harmful online behavior directed at vulnerable individuals and trigger support mechanisms when threatening or distressing content is detected.

In Law Enforcement and Cybercrime Units Cybercrime cells can utilize the framework to gather evidence of online harassment, track repeated offenders, and analyze multilingual abusive behavior across social media channels, thereby accelerating investigations.

Another important application is in online comment moderation systems used in blogs, forums, and news websites. Many websites allow users to post comments under articles or discussion threads. By integrating the cyberbullying detection system into these platforms, harmful comments can be filtered automatically before they become visible to other users. This helps maintain a respectful and constructive discussion environment.

The project can also be used in content moderation tools for digital communities and gaming platforms. Online gaming communities often involve chat-based communication between players, which can sometimes include offensive or abusive language. Integrating cyberbullying detection systems into gaming platforms can help reduce toxic behavior and promote a more positive gaming experience.

Cyberbullying detection systems can also be used in live streaming platforms where users interact through real-time chat. In such platforms, viewers often post comments while watching live videos. By integrating automated detection systems, harmful comments can be blocked instantly before they appear in the chat window, ensuring a safer and more positive interaction between viewers and content creators.

Additionally, the system can be used by organizations and companies to monitor online brand reputation and customer feedback. Businesses can analyze customer comments and identify abusive or harmful language directed toward employees or other customers. This helps organizations maintain professional communication channels and improve their customer interaction policies.

Overall, the cyberbullying detection system developed in this project has wide-ranging applications in social media monitoring, online communication platforms, educational environments, and content moderation systems. By automatically identifying and filtering harmful comments, the system contributes to building safer and more responsible digital communication environments.

In conclusion, CrossTongueGuard offers a scalable and versatile solution that strengthens digital safety across a wide range of social, educational, and technological ecosystems.

Chapter 2

LITERATURE SURVEY

2.1 What is literature survey

A literature survey, also known as a literature review or analysis, plays a pivotal role in various projects, particularly in academic research endeavors. It involves a systematic and thorough examination of existing literature, encompassing research papers, articles, books, and relevant sources pertaining to the project's subject or topic. The central aims of a literature survey within a project encompass gaining in-depth knowledge and comprehension of existing theories, concepts, and research findings, which serve as the project's foundational building blocks. Moreover, it aids in identifying research gaps, unresolved questions, and areas necessitating further exploration, ultimately shaping the project's research objectives and questions.

Furthermore, a literature survey contributes to the development of a theoretical framework, guiding the project's conceptual and analytical aspects. It provides methodological guidance by offering insights into the research methodologies and techniques employed in similar studies, assisting researchers in selecting appropriate approaches for data collection and analysis. By focusing on hybrid architecture, the project ensures that both local patterns and global correlation with physiological data are adequately captured. This process also enables researchers to bolster their arguments and findings by contextualizing them within the existing literature. This Comparative analysis assists in selecting appropriate methodologies that suit the specific requirements of the project, whether qualitative, quantitative, or mixed method approaches.

2.2 Related Work

Cyberbullying detection has attracted significant attention from researchers in the fields of natural language processing, machine learning, and social computing. Several studies have explored different techniques for identifying abusive language and harmful behavior in online communication. These studies have contributed to the development of automated systems capable of detecting cyberbullying in textual data generated on social media platforms.

One of the early approaches to cyberbullying detection focused on the use of traditional machine learning algorithms combined with text-based feature extraction techniques. Researchers applied algorithms such as Naïve Bayes, Support Vector Machines (SVM), and Logistic Regression to classify online comments as abusive or non-abusive. These models relied on textual features extracted using techniques such as Bag-of-Words and Term Frequency–Inverse Document Frequency (TF-IDF). Although these methods provided reasonable results, they often struggled to capture contextual relationships between words within a sentence.

In recent years, transformer-based models have achieved significant improvements in natural language processing tasks. Models such as BERT and its multilingual variants have demonstrated strong performance in tasks such as text classification, sentiment analysis, and hate speech detection. Transformer models use attention mechanisms to capture contextual relationships between words in a sentence, allowing them to understand semantic meaning more effectively than earlier approaches.

Researchers have also explored multilingual transformer models for detecting harmful content in diverse linguistic environments. Models such as MuRIL and XLM-RoBERTa have been specifically designed to handle multilingual text data. These models are capable of processing text from multiple languages and learning cross-lingual representations, making them suitable for analyzing online comments posted by users from different linguistic backgrounds.Transformer models use attention mechanisms to capture contextual relationships between words in a sentence, allowing them to understand semantic meaning more effectively than earlier approaches.

Despite the progress made in previous research, several challenges still remain in cyberbullying detection. Many existing approaches rely on simple binary labeling of comments without considering deeper semantic attributes such as intent, harm, or aggression. Additionally, dataset imbalance and noisy textual data continue to affect the performance of classification models. Addressing these challenges requires improved dataset annotation strategies and more advanced modeling techniques.

In this project, a cyberbullying detection framework is developed by combining semantic context-based annotation with multiple machine learning and transformer-based models. By incorporating additional semantic attributes such as intent, harm, and aggression, the proposed approach aims to improve the accuracy and reliability of cyberbullying detection. The performance of traditional machine learning models is compared with advanced transformer models to identify the most effective approach for detecting harmful online comments.

[1] Maity, K., Jain, R., Jha, P., Saha, S. (2024). Explainable cyberbullying detection in Hinglish: A generative approach. IEEE Transactions on Computational Social Systems, 11(3), 3338–3347.

This study examines the complete explainable cyberbullying detection framework for Hinglish code-mixed social media text by introducing a new dataset, BullyExplain, and a unified generative model named BullyGen. The dataset is the first of its kind to include bully labels, sentiment classes, target categories, and human-annotated rationales, enabling deeper interpretability and richer supervision.The proposed BullyGen model reformulates the multitask problem—cyberbullying detection, rationale extraction, target identification, and sentiment analysis—as a text-to-text generation task, allowing the use of powerful pretrained sequence-to-sequence models like T5 and BART.

Extensive experiments show that BullyGen significantly outperforms traditional baselines and the previous state-of-the-art, improving accuracy, rationale quality, and overall robustness. The study concludes that combining rationales, sentiment, and target information within a generative multitask architecture leads to more accurate, interpretable, and reliable cyberbullying detection in multilingual social media environments.

[2] Khanuja, S., Bansal, D., Mehtani, S., Khosla, S., Dey, A., Gopalan, B., ... Talukdar, P. (2021). Muril: Multilingual representations for indian languages. arXiv preprint arXiv:2103.10730.

This study introduces MuRIL (Multilingual Representations for Indian Languages), a transformer-based language model designed specifically for Indian languages. The authors focused on improving natural language processing performance for languages such as Hindi, Tamil, Telugu, Bengali, and others. They trained the model using large multilingual datasets and included both translated and transliterated text to better capture linguistic variations across Indian languages.

The model was evaluated on several NLP tasks such as text classification, sentiment analysis, and question answering. Results showed that MuRIL significantly improved performance compared with other multilingual models when processing Indian languages and code-mixed text. The study demonstrated that language models trained with region-specific linguistic data can provide more accurate results for multilingual environments.

[3] He, P., Liu, X., Gao, J., Chen, W. (2021). " DeBERTa: Decoding-enhanced BERT with Disentangled Attention". In Proceedings of the 9th International Conference on Learning Representations, ICLR.

In this research, the authors proposed an improved transformer-based language model called DeBERTa to enhance the performance of the BERT architecture. The main idea behind DeBERTa is the disentangled attention mechanism, which separates the content information of a word and its positional information during the attention process. This allows the model to better capture relationships between words and understand the context of sentences more effectively than traditional BERT models.

Additionally, the authors introduced an improved mask decoding strategy to strengthen the pre-training process. They evaluated DeBERTa on several benchmark Natural Language Processing (NLP) tasks such as text classification, question answering, and natural language inference, where it achieved better performance compared to BERT and RoBERTa models.

[4] Conneau, A., Khandelwal, K., Goyal, N., Chaudhary, V., Wenzek, G., Guzmán, F., ... Stoyanov, V. (2020, July). Unsupervised cross-lingual representation learning at scale. In Proceedings of the 58th annual meeting of the association for computational linguistics (pp. 8440-8451).

This paper presents XLM-R (Cross-lingual Language Model-RoBERTa), a large-scale multilingual transformer model designed to support many languages simultaneously. The authors trained the model using massive multilingual datasets collected from various online sources. Their goal was to improve cross-lingual representation learning so that a single model could understand multiple languages effectively.

The model was evaluated on various cross-lingual NLP tasks such as translation, classification, and question answering. Results showed that XLM-R achieved higher accuracy compared with previous multilingual models, especially for low-resource languages. The research demonstrated the effectiveness of large-scale multilingual training for improving natural language understanding across different languages.

[5] Liu, Y., Ott, M., Goyal, N., Du, J., Joshi, M., Chen, D., ... Stoyanov, V. (2019). Roberta: A robustly optimized bert pretraining approach. arXiv preprint arXiv:1907.11692.

This project introduced RoBERTa (Robustly Optimized BERT Pretraining Approach), an improved version of the BERT model. The authors analyzed the training strategy used in BERT and identified several limitations related to data size, training duration, and hyperparameter selection. They proposed modifications to the training process to improve the model's performance without changing the core architecture.

By training the model on larger datasets and using optimized training techniques, RoBERTa achieved better results on many NLP benchmark tasks such as text classification and question answering. These results highlight the importance of previously overlooked design choices, and raise questions about the source of recently reported improvements. We release our models and code.The study showed that careful optimization of the training process can significantly improve the effectiveness of transformer-based language models.

[6] Devlin, J., Chang, M. W., Lee, K., Toutanova, K. (2019, June). Bert: Pre-training of deep bidirectional transformers for language understanding. In Proceedings of the 2019 conference of the North American chapter of the association for computational linguistics: human language technologies, volume 1 (long and short papers) (pp. 4171-4186).

This research introduced BERT (Bidirectional Encoder Representations from Transformers), a revolutionary model for natural language processing. Unlike previous models that read text in a single direction, BERT processes text in both directions simultaneously, enabling it to capture deeper contextual relationships between words.

The model was trained using large text corpora and evaluated on several NLP tasks such as sentiment analysis, question answering, and language inference. BERT achieved state-of-the-art results on many benchmarks and significantly advanced the field of natural language processing. The study laid the foundation for many later models such as RoBERTa, DeBERTa, and multilingual transformer architectures

[7] Davidson, T., Warmsley, D., Macy, M., Weber, I. (2017, May). Automated hate speech detection and the problem of offensive language. In Proceedings of the international AAAI conference on web and social media (Vol. 11, No. 1, pp. 512-515).

In this work, the authors focused on building an automated system to detect hate speech and offensive language in social media, especially on Twitter. They collected a large dataset of tweets and manually labeled them into three categories: hate speech, offensive language, and neither. The researchers then applied Natural Language Processing (NLP) techniques such as text preprocessing, tokenization, and feature extraction to prepare the data for machine learning models.

They trained several machine learning classifiers, including logistic regression and other supervised models, using features like n-grams, TF-IDF values, and linguistic patterns to identify harmful content. One of the key contributions of this study was highlighting the difficulty of distinguishing between hate speech and general offensive language, since many offensive posts are not necessarily hate speech. Their results showed that automated systems can detect abusive content with reasonable accuracy but may still confuse offensive language with true hate speech. This research provided an important dataset and baseline models that later studies used to improve cyberbullying and hate speech detection systems.

[8] Malmasi, S., Zampieri, M. (2017). Detecting hate speech in social media. arXiv preprint arXiv:1712.06427.

The researchers used a labeled dataset containing social media posts categorized into hate speech, offensive language, and non-offensive content. They applied Natural Language Processing (NLP) techniques to preprocess the text, including tokenization, removal of unnecessary symbols, and extraction of linguistic features. Their goal was to train machine learning models that could automatically classify posts containing abusive or hateful language.

To perform the classification, the authors experimented with several supervised machine learning algorithms using features such as character n-grams, word n-grams, and frequency-based text representations. The results showed that these features can effectively help distinguish between hateful and non-hateful content in social media posts. Their study demonstrated that automated hate speech detection systems can support online moderation and cyberbullying prevention, and it also provided useful baseline results for future research in abusive language detection systems.

[9] Badjatiya, P., Gupta, S., Gupta, M., Varma, V. (2017, April). Deep learning for hate speech detection in tweets. In Proceedings of the 26th international conference on World Wide Web companion (pp. 759-760).

This project focused on detecting hate speech in social media posts, particularly on Twitter. The complexity of the natural language constructs makes this task very challenging.Hate speech detection on Twitter is critical for applications like controversial event extraction, building AI chatterbots, content recommendation, and sentiment analysis.The authors explored the use of deep learning models such as convolutional neural networks and long short-term memory networks for classifying tweets containing offensive or hateful language. Their goal was to improve automated moderation of online platforms.

The study compared deep learning models with traditional machine learning techniques like logistic regression and support vector machines. Results showed that neural network models achieved higher accuracy in detecting hate speech. The research demonstrated the effectiveness of deep learning techniques for identifying abusive language in social media.

[10] Salawu, S., He, Y., Lumsden, J. (2017). Approaches to automated detection of cyberbullying: A survey. IEEE Transactions on Affective Computing, 11(1), 3-24.

In this paper, they presented a systematic review of published research (as identified via Scopus, ACM and IEEE Xplore bibliographic databases) on cyberbullying detection approaches. On the basis of our extensive literature review, we categorise existing approaches into 4 main classes, namely supervised learning, lexicon-based, rule-based, and mixed-initiative approaches. Mixed-initiative approaches show promise by combining human insights with machine capabilities.

Many current approaches focus on isolated aspects, overlooking context or behavioral patterns. Supervised learning often achieves higher accuracy but requires substantial annotated data, while lexicon- and rule-based methods are simpler but less adaptable to evolving language and slang. Addressing dataset scarcity and improving context awareness are critical for future progress. In addition, the project compared different feature extraction techniques such as lexical features, syntactic features, semantic features, and sentiment-based features used in cyberbullying detection models. The authors also reviewed different machine learning algorithms.

[11] Talat, Z., Hovy, D. (2016, June). Hateful symbols or hateful people? predictive features for hate speech detection on twitter. In Proceedings of the NAACL student research workshop (pp. 88-93).

This research explored linguistic and contextual features that can help identify hate speech in online messages. The authors collected a dataset of tweets containing sexist and racist content and analyzed the language patterns used in such posts. They focused on identifying features that could help machine learning models detect abusive behavior.

They have provided a list of criteria founded in critical race theory, and use them to annotate a publicly available corpus of more than 16k tweets. And analyzed the impact of various extra-linguistic features in conjunction with character n-grams for hatespeech detection. They also presented a dictionary based the most indicative words in our data. Several classification models were trained using these features to automatically detect hate speech. The results demonstrated that certain linguistic indicators, such as specific words and patterns, are strong predictors of hateful content. The study provided insights into how machine learning can be used to moderate harmful online communication.

[12] Kowalski, R. M., Giumetti, G. W., Schroeder, A. N., Lattanner, M. R. (2014). Bullying in the digital age: a critical review and meta-analysis of cyberbullying research among youth. Psychological bulletin, 140(4), 1073.

This study provided a comprehensive review of cyberbullying in the digital age. The authors analyzed existing research on online harassment, including its causes, psychological effects, and behavioral patterns. The goal was to understand how digital communication platforms contribute to bullying behavior.

The project also discussed possible strategies for prevention and intervention. It emphasized the importance of developing technological tools and educational programs to detect and reduce cyberbullying. The study serves as a foundation for research on automated systems that identify harmful online content.By systematically mapping existing research, this review highlights gaps and serves as a valuable resource for researchers designing more robust cyberbullying detection systems. Advancing research in this area can contribute to safer online environments and mitigate the psychological impact on vulnerable users.

Mixed-effects meta-analysis results indicate that several psychological and behavioral factors are significantly associated with cyberbullying involvement. Among the strongest associations with cyberbullying perpetration were normative beliefs about aggression and moral disengagement. Individuals who believe that aggressive behavior is acceptable or justified are more likely to engage in cyberbullying activities.

[13] Barman, U., Das, A., Wagner, J., Foster, J. (2014, October). Code mixing: A challenge for language identification in the language of social media. In Proceedings of the first workshop on computational approaches to code switching (pp. 13-23).

This project addressed the challenge of processing code-mixed text, where multiple languages appear in the same sentence. The authors studied social media data where users often mix languages such as English and Hindi while communicating online. Their goal was to identify the language of individual words within mixed sentences.

The researchers proposed techniques for language identification using machine learning models and linguistic features. Their approach improved the accuracy of language detection in code-mixed text. This work highlighted the complexity of multilingual communication in social media and the need for specialized NLP techniques.

[14] Walker, C. M. (2014). Cyberbullying redefined: An analysis of intent and repetition. International Journal of Education and Social Science, 1(5), 59-69.

The purpose of this study was to provide a clear and operational definition of bullying behaviors that occur through the use of information and communication technologies (ICTs). With the rapid growth of digital communication platforms such as social media, messaging applications, and online forums, bullying has increasingly shifted into virtual environments. This form of bullying, commonly referred to as cyberbullying, involves harmful behaviors carried out through electronic devices and internet-based platforms.

Furthermore, the study revealed that cyberbullying behaviors are often more complex and multifaceted than traditional bullying, given the digital context where anonymity, reach, and permanence intensify their impact. Participants reported experiencing a wide range of emotional responses, including anxiety, depression, and feelings of social isolation, emphasizing the psychological toll of online aggression.The paper emphasizes that cyberbullying should not only be identified based on offensive language but also on the intention behind the message and whether the behavior occurs repeatedly.

[15] Chen, Y., Zhou, Y., Zhu, S., Xu, H. (2012, September). Detecting offensive language in social media to protect adolescent online safety. In 2012 international conference on privacy, security, risk and trust and 2012 international confernece on social computing (pp. 71-80). Ieee.

The main objective of the project is that they bulit a system that could analyze user-generated content and classify whether a message contains offensive language. They collected social media text data and applied Natural Language Processing (NLP) techniques to preprocess the text, such as tokenization, removal of stop words, and feature extraction. These processed features were then used to train machine learning classifiers capable of recognizing abusive patterns in online conversations.

The researchers experimented with different text classification approaches to distinguish between normal and offensive content. They mainly relied on linguistic features such as word frequency, n-grams, and contextual patterns to detect harmful expressions. Their system demonstrated that machine learning models can effectively identify offensive language in social media, even when users employ informal writing styles or slang. This work contributed to early research in automated moderation systems, helping platforms detect abusive or harmful messages and laying the foundation for later deep-learning-based hate speech and cyberbullying detection models.

[16] Patchin, J. W., Hinduja, S. (2010). Cyberbullying and self-esteem. Journal of school health, 80(12), 614-621.

This research examined the relationship between cyberbullying and the psychological well-being of students. The authors conducted surveys to study how online harassment affects self-esteem and emotional health. Their findings showed that victims of cyberbullying often experience lower self-confidence and higher stress levels.

The study highlighted the growing problem of cyberbullying in online environments and the need for effective prevention strategies. The authors emphasized that early detection and intervention are important for protecting users from harmful online interactions. This research motivated the development of automated systems for identifying abusive behavior on digital platforms.

2.3 Existing System

With the rapid growth of social media platforms, cyberbullying has become a major issue affecting individuals, especially teenagers and young users. Many existing systems have been developed to detect cyberbullying and offensive language automatically using Natural Language Processing (NLP) and Machine Learning techniques. These systems analyze user-generated content such as comments, tweets, messages, and posts to identify harmful or abusive language. The main goal of these systems is to reduce online harassment and create a safer digital environment.

The existing systems for cyberbullying detection face significant challenges, particularly when handling multilingual and code-mixed communication. Most traditional models rely on surface-level features, which makes them highly sensitive to spelling variations, transliteration differences, and informal grammar commonly present in social media conversations. These models struggle to capture the deeper semantic meaning of mixed-language text, especially when users switch between scripts such as Telugu–English or Hindi–English within the same sentence. Another major limitation is that existing systems treat cyberbullying as a single-label classification problem, focusing only on offensive or hateful content while ignoring crucial behavioral cues such as repetition, intent to harm, and conversational context. Furthermore, many existing datasets are small, language-restricted, or lack proper annotations for complex cyberbullying behaviours, making it difficult for models to generalize across diverse linguistic environments.

Even advanced multilingual transformers suffer performance degradation when trained on limited, noisy, or highly unbalanced datasets. Additionally, the absence of robust preprocessing pipelines for handling code-mixing, script normalization, and profanity variation further reduces accuracy. Overall, current systems are not scalable, not language-agnostic, and fail to provide a comprehensive solution for detecting subtle, evolving, and cross-lingual forms of cyberbullying on global social media platforms.Furthermore, they lack the semantic and contextual understanding required to interpret informal communication styles such as slang, emoji usage, or culturally influenced expressions.

In the early stages of research, cyberbullying detection systems mainly relied on keyword-based methods. These systems used predefined lists of abusive or offensive words to identify harmful messages. If a message contained any word from the predefined list, it was classified as offensive or abusive. Although this approach was simple and easy to implement, it had several limitations. It could not detect cyberbullying when harmful messages were written without explicit offensive words. It also failed to understand sarcasm, indirect insults, or contextual meanings of sentences.

To overcome these limitations, machine learning techniques were introduced in cyberbullying detection systems. These systems used labeled datasets to train classification models that could automatically learn patterns of abusive language. Different algorithms such as Logistic Regression, Naïve Bayes, Support Vector Machines (SVM), and Decision Trees were commonly used for text classification tasks. These models analyzed textual features such as word frequency, n-grams, and sentence structure to classify messages into different categories like cyberbullying, offensive language, or normal content.

Feature extraction played an important role in improving the performance of these machine learning systems. Various linguistic and semantic features were used to better understand the text. Some of the commonly used features included term frequency, sentiment scores, part-of-speech tags, and syntactic patterns. Sentiment analysis was also applied to identify negative emotions in messages. Messages containing strong negative sentiment were often considered potential cyberbullying content.

Later, researchers introduced hybrid approaches that combined rule-based techniques with machine learning models. These hybrid systems attempted to improve detection accuracy by using both predefined rules and learned patterns from datasets. Rule-based components helped detect obvious abusive expressions, while machine learning models handled complex sentence structures and contextual meanings.

As research progressed, attention was given to the context of conversations rather than analyzing individual messages alone. Some cyberbullying behaviors occur through repeated messages or continuous harassment over time. Therefore, some systems began analyzing conversation threads and interaction history between users. This helped identify patterns of repeated harmful behavior, which is an important characteristic of cyberbullying.

Another important aspect studied in existing systems is the intention behind online messages. Not every negative comment can be considered cyberbullying. Some messages may appear offensive but may not actually have harmful intent. Therefore, certain research studies focused on understanding the intent and repetition involved in cyberbullying behavior. These approaches attempted to differentiate between casual online arguments and actual cyberbullying incidents.

With the advancement of Natural Language Processing, deep learning techniques have also been applied in cyberbullying detection systems. Deep learning models such as Recurrent Neural Networks (RNN), Long Short-Term Memory (LSTM), and Convolutional Neural Networks (CNN) have been used to analyze large volumes of text data. These models can automatically learn complex linguistic patterns and contextual relationships between words. Compared to traditional machine learning algorithms, deep learning models often achieve higher accuracy in text classification tasks.

Sarcasm and humor also create difficulties for automated systems. Some messages may appear offensive but are actually jokes among friends, while others may look harmless but carry hidden abusive meanings. Understanding these subtle differences requires deeper contextual and cultural knowledge, which remains a challenge for many automated detection systems.

In summary, existing systems for cyberbullying detection have evolved from simple keyword-based approaches to advanced machine learning and deep learning models. These systems analyze textual features, sentiment patterns, contextual information, and conversation history to identify harmful online behavior. Although significant progress has been made, there is still a need for more intelligent and adaptive systems that can better understand context, intent, and multilingual communication in online environments.

Table 2.1: Related Work

S.No	Title	Authors	Year	Summary
1	Explainable Cyberbullying Detection in Hinglish: A Generative Approach	Maity, K., Jain, R., Jha, P., & Saha, S.	2024	Introduces an explainable cyberbullying detection framework for Hinglish text using a multitask generative model that performs sentiment analysis, rationale extraction and target identification.
2	Multilingual Representations for Indian Languages (MuRIL)	Khanuja, S., Bansal, D., Mehtani, S., Khosla, S., Dey, A., Gopalan, B., & Talukdar, P.	2021	Presents a multilingual transformer model designed for Indian languages, trained with translated and transliterated datasets to improve performance in multilingual NLP tasks.
3	DeBERTa: Decoding-Enhanced BERT with Disentangled Attention	He, P., Liu, X., Gao, J., & Chen, W.	2021	Introduces DeBERTa, an improved transformer architecture that separates content and positional information using disentangled attention to enhance contextual understanding.
4	Unsupervised Cross-Lingual Representation Learning at Scale	Conneau, A., Khandelwal, K., Goyal, N., Chaudhary, V., Wenzek, G., Guzmán, F., & Stoyanov, V.	2020	Presents XLM-R, a multilingual transformer model trained on large multilingual corpora to support cross-lingual tasks such as classification and translation.

S.No	Title	Authors	Year	Summary
5	RoBERTa: A Robustly Optimized BERT Pretraining Approach	Liu, Y., Ott, M., Goyal, N., Du, J., Joshi, M., Chen, D., & Levy, O.	2019	Proposes an optimized version of BERT with improved training strategies and larger datasets, achieving better results in multiple NLP benchmarks.
6	BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding	Devlin, J., Chang, M., Lee, K., & Toutanova, K.	2019	Introduces BERT, a bidirectional transformer model that captures contextual word representations and significantly improves NLP tasks such as sentiment analysis and text classification.
7	Automated Hate Speech Detection and the Problem of Offensive Language	Davidson, T., Warmsley, D., Macy, M., & Weber, I.	2017	Develops a machine learning system to classify tweets into hate speech, offensive language and neutral content using NLP preprocessing and classification algorithms.
8	Detecting Hate Speech in Social Media	Malmasi, S., & Zampieri, M.	2017	Applies supervised machine learning techniques with linguistic features such as n-grams to classify social media posts containing hate speech or offensive language.

S.No	Title	Authors	Year	Summary
9	Deep Learning for Hate Speech Detection in Tweets	Badjatiya, P., Gupta, S., Gupta, M., & Varma, V.	2017	Explores deep learning models such as CNN and LSTM for identifying hate speech in tweets and compares their performance with traditional machine learning methods.
10	Approaches to Automated Detection of Cyberbullying: A Survey	Salawu, S., He, Y., & Lumsden, J.	2017	Provides a comprehensive survey of cyberbullying detection approaches including rule-based, supervised learning and hybrid methods, highlighting key challenges.
11	Hateful Symbols or Hateful People? Predictive Features for Hate Speech Detection on Twitter	Talat, Z., & Hovy, D.	2016	Investigates linguistic and contextual features that help identify hate speech in Twitter data using annotated datasets and machine learning models.
12	Bullying in the Digital Age: A Critical Review and Meta-analysis of Cyberbullying Research Among Youth	Kowalski, R., Giumetti, G., Schroeder, A., & Lattanner, M.	2014	Conducts a meta-analysis of cyberbullying research focusing on psychological effects, causes and behavioral patterns among young social media users.
13	Code Mixing: A Challenge for Language Identification in the Language of Social Media	Barman, U., Das, A., Wagner, J., & Foster, J.	2014	Examines the problem of code-mixed language in social media and proposes machine learning techniques to identify languages at the word level.

S.No	Title	Authors	Year	Summary
14	Cyberbullying Redefined: An Analysis of Intent and Repetition	Walker, C. M.	2014	Analyzes cyberbullying behaviors by emphasizing two major characteristics: intentional harm and repeated aggression in online communication.
15	Detecting Offensive Language in Social Media to Protect Adolescent Online Safety	Chen, Y., Zhou, Y., Zhu, S., & Xu, H.	2012	Proposes an early offensive language detection system using NLP preprocessing and machine learning classifiers based on textual features.
16	Cyberbullying and Self-Esteem	Patchin, J. W., & Hinduja, S.	2010	Examines the psychological effects of cyberbullying and shows that victims often experience reduced self-esteem, stress and emotional distress.

2.4 Proposed System

The proposed system, CrossTongueGuard, introduces a unified and scalable framework designed to detect cyberbullying in multilingual and code-mixed social media conversations. Unlike traditional models that rely solely on aggression or lexical features, the proposed system incorporates a multi-aspect learning architecture that jointly analyses aggression, intent to harm, and repetition patterns, enabling a deeper understanding of cyberbullying behavior. The framework integrates machine learning models such as Logistic Regression, Support Vector Machine, Naive Bayes and advanced multilingual transformer models such as MuRIL, XLM-RoBERTa, and mDeBERTa, which are capable of handling diverse scripts, transliterated text, noisy user-generated content and generates accurate results by predicting whether the given text is cyberbullying or not. This approach helps in effectively identifying harmful communication patterns across multiple languages and social media platforms.

The core components of CrossTongueGuard are structured to ensure end-to-end multilingual cyberbullying detection with high reliability. The system begins with a Multilingual Preprocessing Module, which performs token cleaning, normalization, transliteration correction, code-switch identification, and script unification. This ensures uniform text representation across mixed-script content. Next, the cleaned input is passed into the Transformer-based Multilingual Encoder, which may consist of machine learning models like Logistic Regression,Support Vector Machine,Naive Bayes and transformer models MuRIL, XLM-RoBERTa, or mDeBERTa depending on the language pair and model configuration. These encoders extract deep contextual embeddings that capture sentiment, tone, sarcasm, and hidden meanings prevalent in social media conversations.

It aims to develop an intelligent framework for detecting cyberbullying in social media text using advanced Natural Language Processing (NLP) and machine learning techniques. With the rapid expansion of social networking platforms, millions of users interact daily through comments, posts, and private messages. Although these platforms provide a convenient way for communication and information sharing, they also create opportunities for negative behaviors such as harassment, hate speech, and cyberbullying. Cyberbullying can have serious psychological and emotional impacts on individuals, especially teenagers and young users. Therefore, it is important to design automated systems that can detect harmful online behavior quickly and efficiently. The proposed system focuses on analyzing user-generated textual content and identifying whether the message contains cyberbullying behavior or not.

The system is designed to process textual data from social media platforms and classify it into two categories: cyberbullying and non-cyberbullying. Unlike traditional rule-based methods that rely mainly on keyword matching, the proposed system uses advanced language models capable of understanding the context and meaning of the text. Cyberbullying messages are often expressed indirectly through sarcasm or subtle insults, which makes them difficult to detect using simple keyword-based approaches. By using modern NLP techniques, the proposed system can capture deeper linguistic patterns and contextual relationships between words.

The first stage of the proposed system involves data collection and preprocessing. Social media text is usually unstructured and contains various types of noise such as emojis, URLs, hashtags, user mentions, and repeated characters. These elements may affect the performance of machine learning algorithms if not handled properly. Therefore, preprocessing is performed to clean and normalize the text data. This step includes converting all characters to lowercase, removing unnecessary punctuation marks, eliminating hyperlinks, and filtering out irrelevant symbols. Tokenization is then applied to divide the text into individual words or tokens.

Stop words such as common articles, pronouns, and conjunctions are removed since they do not contribute significantly to the meaning of the sentence. These preprocessing steps help transform raw text into a structured format that can be effectively analyzed by machine learning models.The preprocessed data is ready for tranforming into numerical representations.

The text data is transformed into numerical representations through feature extraction techniques. Computers cannot directly interpret textual information, so words must be represented as numerical vectors before they can be processed by algorithms. Traditional feature extraction methods such as bag-of-words and TF-IDF represent text based on word frequency. However, these approaches often fail to capture contextual meaning and semantic relationships between words.

To overcome this limitation, the proposed system utilizes advanced word embedding techniques and transformer-based language models. These models generate contextual representations of words by considering their surrounding context in a sentence. As a result, the system can better understand the meaning of complex expressions and detect abusive language even when it appears in indirect or subtle forms.

The core component of the proposed system is the classification model responsible for identifying cyberbullying content. This model is trained using a labeled dataset containing examples of both offensive and non-offensive messages. During the training process, the model learns patterns and features associated with harmful language, personal attacks, threats, and harassment.

Deep learning models are particularly effective in this context because they can automatically learn complex patterns from large datasets without requiring manual feature engineering. Once the model has been trained, it can analyze new messages and predict whether they contain cyberbullying behavior.

Another important aspect considered in the proposed system is the behavioral characteristics of cyberbullying. Cyberbullying is not always limited to a single offensive message. In many cases, it involves repeated harassment or targeted attacks against a specific individual over a period of time. By analyzing patterns of repeated messages or aggressive interactions within a conversation, the system can better identify instances of cyberbullying. This helps distinguish between casual online arguments and actual bullying behavior. Incorporating behavioral patterns into the detection process enhances the overall reliability of the system.

The proposed system is also designed to handle multilingual and code-mixed text, which is common in social media communication. Users frequently combine multiple languages within the same message, such as mixing English with regional languages. Traditional language processing systems often struggle to analyze such mixed-language content effectively. By using multilingual language models, the proposed system can process and understand text written in different languages or combinations of languages. This capability makes the system more adaptable to real-world social media environments where linguistic diversity is common.

Finally, the system produces classification results indicating whether a given message contains cyberbullying content. These predictions can be used by social media platforms, online communities, or moderation systems to identify harmful messages and take appropriate actions. For example, suspicious messages may be flagged for review by moderators, automatically filtered, or used to warn users about inappropriate behavior. By integrating advanced language models, contextual text analysis, and behavioral pattern recognition, the proposed system provides a more accurate and efficient solution for cyberbullying detection. Implementing such automated systems can significantly reduce harmful online interactions and contribute to creating a safer and more supportive digital environment for users.

Chapter 3

DESIGN

The design of the proposed cyberbullying detection system focuses on developing an efficient framework that can automatically analyze social media text and identify harmful or abusive content. The system is designed to process user-generated messages and classify them into cyberbullying or non-cyberbullying categories. Since social media platforms generate a huge volume of textual data every day, manual monitoring becomes difficult and inefficient. Therefore, the system design emphasizes automation, accuracy, and scalability so that harmful content can be detected quickly. The overall design integrates Natural Language Processing (NLP) techniques with transformer-based deep learning models to understand the context and meaning of social media text.

The first component in the system design is the data input and collection stage. In this stage, textual data is gathered from social media sources such as comments, posts, tweets, or chat messages. The collected dataset includes examples of both normal conversations and cyberbullying messages. Since the data obtained from online platforms is usually unstructured and contains many informal expressions, it requires proper handling before it can be analyzed. The system design ensures that the input module accepts text data and forwards it to the preprocessing stage for cleaning and preparation.

The preprocessing stage plays a crucial role in improving the quality of the input data. Social media messages often contain noise such as emojis, hyperlinks, hashtags, mentions, and irregular spellings. These elements may reduce the performance of machine learning models if they are not processed properly. Therefore, several preprocessing steps are applied, including converting text into lowercase, removing punctuation marks, eliminating URLs, and filtering out unnecessary symbols. Tokenization is then used to divide the sentences into individual words or tokens. Stop words that do not contribute significantly to the meaning of the sentence are removed. These preprocessing operations help in transforming raw text into a structured format suitable for further analysis.

After preprocessing, the next important stage in the system design is feature extraction. In this stage, the cleaned text is converted into numerical representations that machine learning models can understand. Traditional methods such as bag-of-words or TF-IDF represent text based on word frequency, but they do not capture contextual meaning effectively. To address this limitation, the proposed system utilizes transformer-based language models that generate contextual embeddings for words. These models analyze the relationship between words in a sentence and capture deeper semantic meaning. This helps the system understand complex expressions, sarcasm, and indirect insults that are often present in cyberbullying messages.

The classification module is the core component of the system design. In this stage, the numerical features generated during feature extraction are passed to a deep learning model trained to detect cyberbullying patterns. The model learns from labeled datasets containing examples of abusive and non-abusive messages. During the training process, it identifies patterns related to insults, threats, harassment, and offensive language. Once the model is trained, it can analyze new input messages and predict whether they contain cyberbullying behavior. This classification process enables the system to automatically detect harmful content in social media text.

Another important aspect of the system design is its ability to handle multilingual and code-mixed text. Many social media users combine different languages within a single sentence, such as mixing English with regional languages. Traditional language processing systems often struggle with such mixed language patterns. By incorporating multilingual language models, the system can analyze text written in multiple languages or combinations of languages. This improves the system's ability to detect cyberbullying across diverse online communities. The input text is analyzed and perform preprocessing before predicting output.

Finally, the output module displays the classification results generated by the system. If a message is identified as cyberbullying, the system blocks the message and it will not be dispalyed. These results can then be used by social media platforms or moderation systems to take appropriate action, such as filtering the message, warning the user, or sending it for moderator review. Through this structured design, the proposed system provides an effective approach for detecting cyberbullying and promoting safer online communication environments.

3.1 Software Requirements

The development and implementation of the CrossTongueGuard require several software tools and technologies to ensure efficient data processing, model training, and system deployment. These software components support various stages of the project, including data preprocessing, feature extraction, model development, and evaluation. Proper selection of software tools is important for building an effective and reliable system that can handle large amounts of textual data from social media platforms.

The primary programming language used for implementing the proposed system is Python. Python is widely used in the field of machine learning and Natural Language Processing due to its simplicity, flexibility, and large number of available libraries. It provides powerful frameworks and tools that simplify the process of text analysis, data manipulation, and deep learning model development. Python also supports integration with several machine learning libraries that are useful for training and testing cyberbullying detection models.

Several important Python libraries are used in the development of the system. Libraries such as NumPy and Pandas are used for data handling and preprocessing tasks. These libraries help in managing large datasets, performing data cleaning, and organizing textual data for further processing. For Natural Language Processing tasks, libraries such as NLTK and SpaCy are used to perform operations like tokenization, stop-word removal, and text normalization. These tools help in preparing the raw text data for feature extraction and model training.

The system also utilizes deep learning frameworks such as TensorFlow and PyTorch to build and train machine learning models. These frameworks provide efficient tools for implementing neural networks and transformer-based architectures used for cyberbullying detection. They support advanced features such as automatic differentiation, GPU acceleration, and scalable training, which help improve model performance and reduce training time. Pre-trained transformer models can also be integrated using libraries such as Hugging Face Transformers, which provide easy access to modern language models.

3.2 Hardware Requirements

The development and implementation of the CrossTongueGuard require certain hardware components to ensure smooth execution of data processing, model training, and system testing. Since the project involves Natural Language Processing (NLP) and deep learning models, the hardware should be capable of handling large datasets and performing computational tasks efficiently. Proper hardware resources help in reducing training time, improving system performance, and enabling faster processing of social media text data.

The most important hardware component required for the system is a computer or laptop with a reliable processor. A system with at least an Intel Core i5 processor or an equivalent processor is recommended to efficiently run machine learning algorithms and data preprocessing tasks. The processor is responsible for executing program instructions and handling complex computations involved in text analysis and model training. A higher processing speed allows the system to process large volumes of social media data more efficiently.

Another important requirement is sufficient memory (RAM). A minimum of 8 GB RAM is recommended for the development of the cyberbullying detection system. RAM plays a crucial role in handling datasets and executing machine learning models during training and testing. When working with large text datasets and deep learning frameworks, higher memory capacity ensures that the system can process data smoothly without slowing down or crashing. For better performance, systems with 16 GB RAM can provide faster model training and improved multitasking capabilities.

Storage is also an essential hardware requirement for the project. The system should have at least 256 GB of storage space to store datasets, machine learning libraries, program files, and trained models. Solid State Drives (SSD) are preferred over traditional Hard Disk Drives (HDD) because they provide faster data access and improve overall system performance. Adequate storage space ensures that large datasets and model outputs can be stored and accessed efficiently.

3.3 Architecture

The proposed system architecture describes the overall workflow used to detect cyberbullying in code-mixed social media text. The system processes the input text through multiple stages including preprocessing, labeling, feature extraction, model training, and prediction. Each stage performs a specific function to transform raw social media data into meaningful outputs that can determine whether a message contains cyberbullying content.

Figure 3.1 illustrates the overall architecture of the CrossTongueGuard system. The system begins by accepting code-mixed social media text as input. Social media messages often contain informal language, multiple languages, emojis, and special characters. Therefore, the first step in the architecture is the text preprocessing module, where the input text is cleaned and normalized. This stage includes cleaning unwanted characters, normalizing the text format, and removing noise such as URLs, emojis, and irrelevant symbols. These preprocessing operations ensure that the text data becomes structured and suitable for further analysis.

After preprocessing, the system performs semantic context-based labeling. In this stage, the system analyzes the contextual meaning of the text and assigns appropriate labels. Instead of relying only on keywords, the system evaluates the surrounding context of words to identify cyberbullying behavior. Binary labels are assigned to classify the messages as either cyberbullying or non-cyberbullying. This stage improves the system's ability to detect indirect insults, sarcasm, and aggressive expressions commonly found in social media communication.

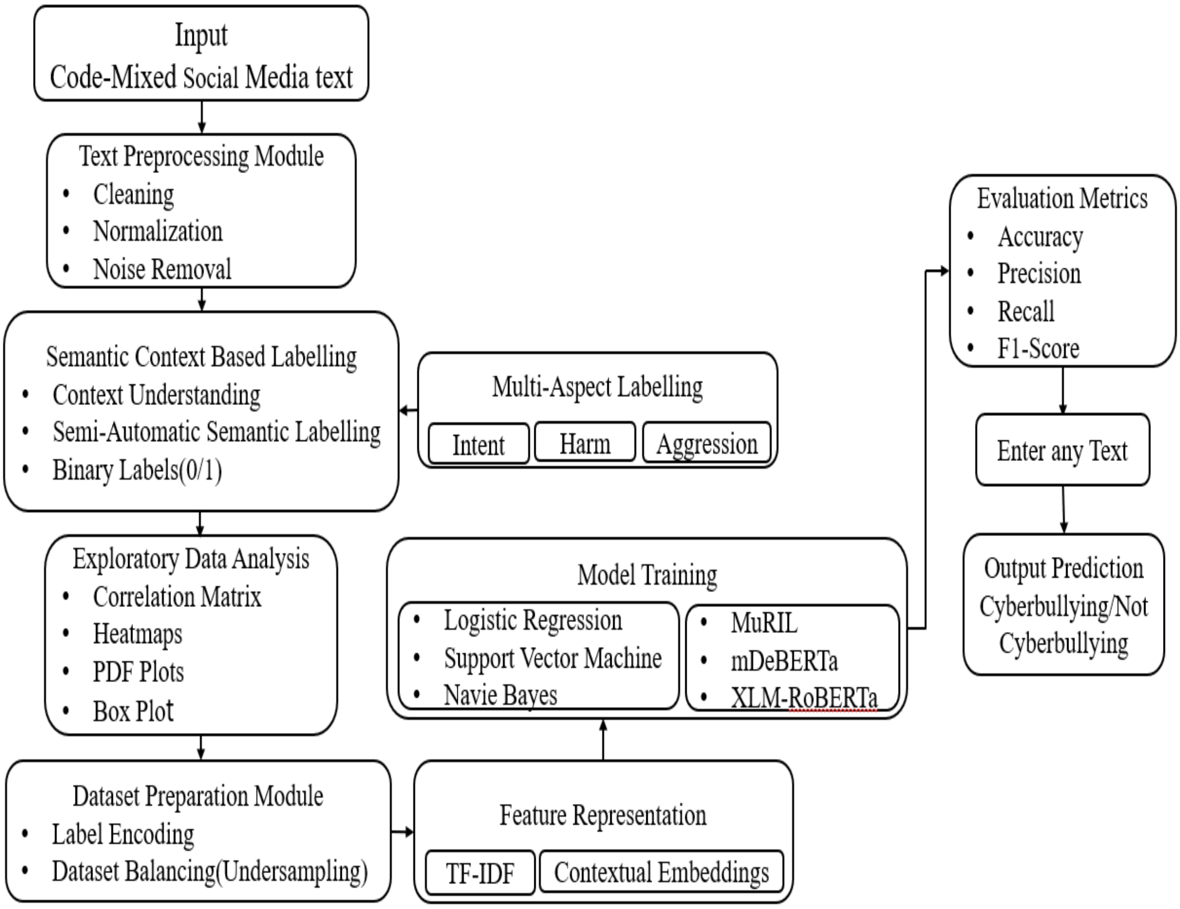

Figure 3.1: Architecture of CrossTongueGuard

The architecture also includes a multi-aspect labeling component that examines different behavioral characteristics of cyberbullying. These aspects include intent, harm, and aggression. The intent aspect determines whether the message is deliberately intended to harm someone. The harm aspect evaluates the emotional impact of the message on the target individual, while the aggression aspect identifies the presence of abusive or hostile language. Considering multiple aspects allows the system to understand cyberbullying behavior more effectively

Next, exploratory data analysis is performed to understand the patterns and relationships present in the dataset. Visualization techniques such as correlation matrices, heatmaps, probability distribution plots, and box plots are used to analyze the data. This stage helps identify important features and patterns that contribute to cyberbullying detection. The dataset preparation module then prepares the labeled data for model training. In this stage, label encoding is applied to convert categorical labels into numerical values that can be processed by machine learning models. Dataset balancing techniques such as undersampling are also applied to ensure that both cyberbullying and non-cyberbullying classes are represented equally in the dataset. Balanced datasets help improve the reliability and fairness of the trained models.

After dataset preparation, feature representation techniques are applied to convert textual data into numerical feature vectors. Methods such as TF-IDF are used to represent word importance within the dataset. In addition, contextual embeddings are generated using transformer-based language models, which capture deeper semantic relationships between words and phrases.

The extracted features are then used for model training. The system uses both traditional machine learning algorithms and advanced transformer-based models for classification. Traditional models such as Logistic Regression, Support Vector Machine, and Naive Bayes provide baseline performance, while transformer models such as MuRIL, mDeBERTa, and XLM-RoBERTa capture contextual and multilingual information from the text.

Finally, the performance of the trained models is evaluated using metrics such as accuracy, precision, recall, and F1-score. These metrics help determine how effectively the model can identify cyberbullying messages. After evaluation, the system allows users to enter any new text input, and the trained model predicts whether the message represents cyberbullying or not. The final output provides a clear classification that can be used for monitoring and moderating harmful online communication.

Figure 3.2 illustrates the architecture of the Streamlit-based user interface designed for the cyberbullying detection system. The interface acts as the front-end component that allows users to interact with the trained machine learning model. Through the Streamlit interface, users can enter any message or social media text that needs to be analyzed. Once the user provides the input, the system loads the trained model and sends the text to the inference engine for processing. The inference engine performs text preprocessing operations such as cleaning the text and preparing it for analysis. It then generates contextual embeddings that help the model understand the meaning and context of the message rather than simply relying on individual words.

After the inference process is completed, the trained model produces a prediction indicating whether the message contains cyberbullying content or not. Based on the prediction result, the system performs a decision step to determine the appropriate action. If the message is classified as cyberbullying, the system blocks the message and prevents it from being delivered. On the other hand, if the message does not contain cyberbullying content, it is allowed to be delivered normally. This architecture demonstrates how the cyberbullying detection model can be integrated with a user-friendly interface to analyze messages in real time and help prevent the spread of harmful or abusive communication on online platforms

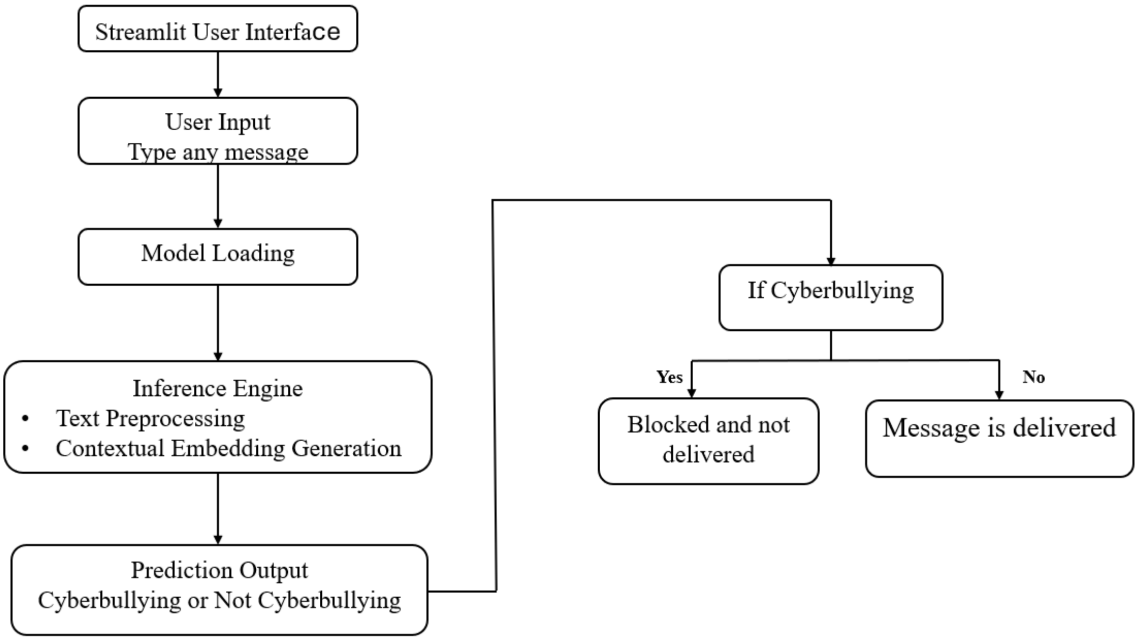

Figure 3.2: User Interface Architecture for CrossTongueGuard

3.4 UML Diagrams

Unified Modeling Language (UML) is a standardized modeling language used to visually represent the design and structure of software systems. It provides a set of diagrams that help developers understand, design, and document the architecture and behavior of a system. UML diagrams simplify complex system structures by representing components, relationships, and interactions in a graphical format. This makes it easier for developers, researchers, and stakeholders to understand how different parts of the system work together.

In software development projects, UML diagrams help in planning the system before actual implementation. They allow developers to visualize system components, workflows, and interactions between different modules. By using UML diagrams, it becomes easier to identify system requirements, detect potential design issues, and improve overall system organization. In the proposed cyberbullying detection system, UML diagrams are used to represent different aspects of the system such as user interaction, system workflow, structural relationships between components, and communication between modules. The main UML diagrams used in this project include the Use Case Diagram, Class Diagram, Activity Diagram, and Sequence Diagram.

3.4.1 Class Diagram

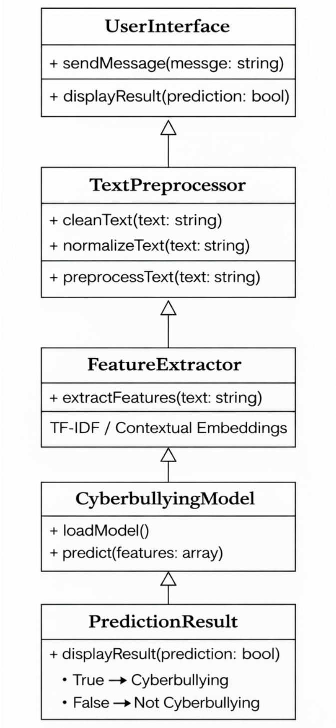

Figure 3.3: Class Diagram

The class diagram represents the structural design of the cyberbullying detection system by showing the classes, attributes, methods, and relationships between different components. The system consists of several classes such as User Interface, Text Preprocessor, Feature Extractor, Cyberbullying Detection Model, and Prediction Result. Each class performs a specific function in processing the input message and generating the final prediction.

The User Interface class handles user interaction by receiving text input and displaying prediction results. The Text Preprocessor class cleans and normalizes the input text by removing noise such as special characters and URLs. The Feature Extractor class converts the processed text into numerical feature representations such as TF-IDF or contextual embeddings. The Cyberbullying Detection Model class analyzes these features and predicts whether the message contains cyberbullying content. Finally, the Prediction Result class displays the classification output to the user. The class diagram helps organize system components and shows how they are connected.

3.4.2 Use Case Diagram

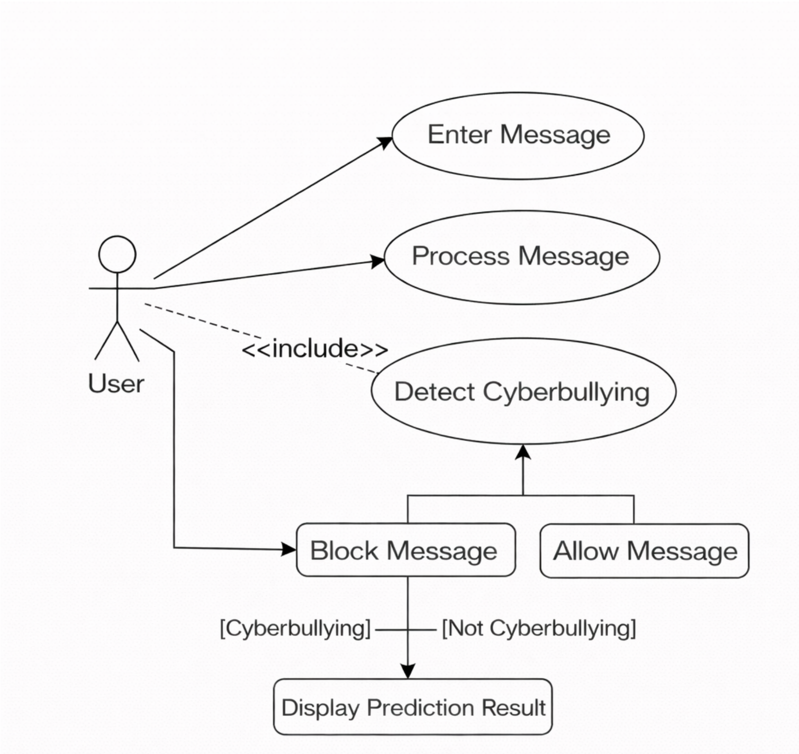

Figure 3.4: Use Case Diagram

The use case diagram represents the interaction between users and the cyberbullying detection system. It illustrates how the user interacts with the system to perform specific operations. In the proposed system, the main actor is the user who enters a message through the user interface. The system then processes the input message and performs cyberbullying detection using the trained model. The use case diagram helps identify the main functionalities provided by the system and how users access them.

The major use cases in this system include entering text input, processing the message, analyzing the text using the trained machine learning model, and displaying the prediction result. The system evaluates the message and determines whether it contains cyberbullying content or not. If the message is detected as cyberbullying, the system blocks the message; otherwise, the message is delivered normally. The use case diagram helps in understanding the overall functionality of the system from the user's perspective.

3.4.3 Sequence Diagram

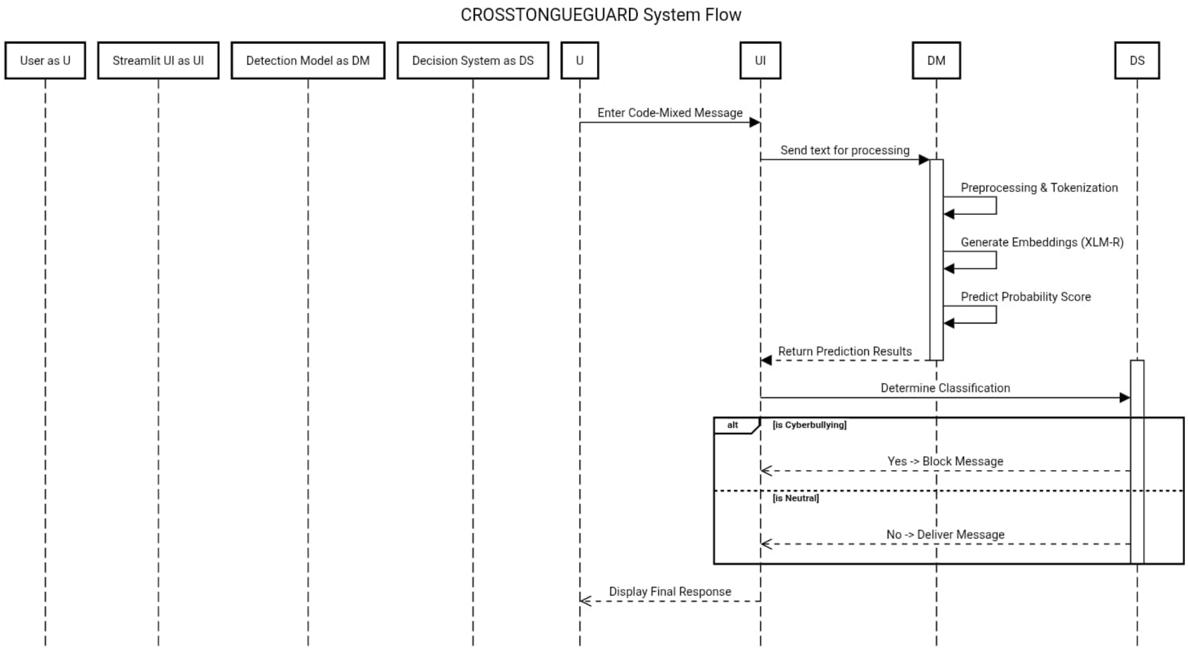

Figure 3.5: Sequence Diagram

The sequence diagram represents the interaction between different components of the cyberbullying detection system in a time-ordered manner. The process begins when the user enters a message through the Streamlit user interface. The interface captures the input and sends the text to the preprocessing module. In this stage, the system performs several text preparation steps such as cleaning unwanted characters, removing stop words, and converting the message into a format suitable for machine learning analysis. This prepared text is then forwarded to the trained machine learning model for prediction.After receiving the processed input, the machine learning model analyzes the content using the trained cyberbullying detection algorithm.

The model generates a prediction indicating whether the message contains cyberbullying content or not. The result is then sent to the output system, which determines the final action. If the message is identified as cyberbullying, the system blocks the message and prevents it from being delivered. Otherwise, the message is allowed to pass through normally. Finally, the result is displayed back to the user through the Streamlit interface, completing the interaction flow of the system.

3.4.4 Activity Diagram

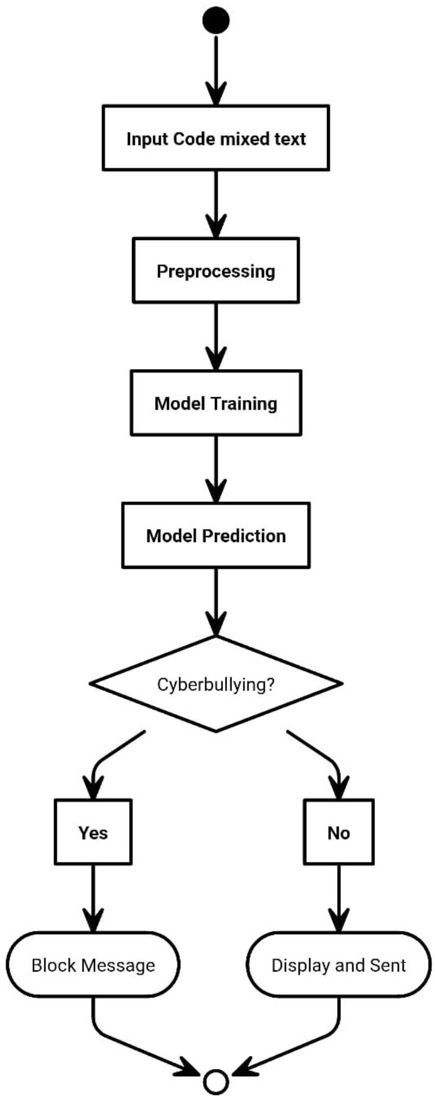

Figure 3.6: Activity Diagram

The activity diagram represents the workflow of the cyberbullying detection system. It shows the sequence of activities performed by the system from the moment a user enters a message until the final prediction is generated. The process begins when the user enters a message through the user interface. The system then preprocesses the input text by cleaning and normalizing it to remove unnecessary elements.

After preprocessing, feature extraction is performed to convert the text into numerical representations. The trained machine learning model then analyzes the extracted features to determine whether the message contains cyberbullying content. If the message is classified as cyberbullying, the system blocks the message; otherwise, it allows the message to be delivered. The activity diagram clearly illustrates the flow of operations and decision points within the system.

Chapter 4

IMPLEMENTATION

The implementation phase is one of the most important stages in the development of the cyberbullying detection system, as it focuses on transforming the system design into a fully functional application. In this project, the implementation is carried out using the Python programming language because of its strong ecosystem for machine learning, natural language processing, and web application development. Various Python libraries and frameworks are used to support the development process. The system integrates a machine learning model with an interactive user interface to provide real-time detection of cyberbullying messages. The overall goal of the implementation is to ensure that the system can efficiently process user input, analyze the text using the trained model, and produce an accurate prediction regarding whether the message contains cyberbullying content or not.

The user interface of the system is implemented using the Streamlit framework. Streamlit is a lightweight and efficient Python library that allows developers to build interactive web applications with minimal effort. Through this interface, users can easily type or paste a message into the input field provided on the screen. The interface is designed to be simple and user-friendly so that even users with minimal technical knowledge can interact with the system without difficulty. Once the application is launched, the trained machine learning model is loaded into memory. Loading the model at the beginning ensures that predictions can be generated quickly whenever a user submits a message for analysis. The Streamlit interface acts as the bridge between the user and the underlying machine learning system.

After the user enters a message, the system begins the text processing phase. Text preprocessing is an essential step in natural language processing because raw text data often contains unnecessary characters, formatting inconsistencies, and other elements that may affect the performance of the machine learning model. During preprocessing, the system performs operations such as converting all characters to lowercase, removing punctuation marks, eliminating special characters, and cleaning unwanted spaces. These preprocessing steps help standardize the text and make it easier for the model to analyze patterns within the message. In some cases, additional preprocessing techniques such as tokenization or stop-word removal may also be applied depending on the dataset and model requirements.

Once the text is cleaned and prepared, the processed input is passed to the inference engine of the system. The inference engine is responsible for generating contextual representations of the input text. In this project, contextual embeddings are generated to capture the semantic meaning of the message. Unlike traditional methods that analyze words individually, contextual embeddings help the model understand the relationships between words and the context in which they appear. This improves the system's ability to detect subtle forms of cyberbullying that may not rely on explicit offensive words but still convey harmful intent. The generated embeddings are then passed to the trained classification model for prediction.

The machine learning model used in the system has been previously trained on a dataset containing examples of both cyberbullying and not cyberbullying messages. During the training phase, the model learns patterns and linguistic features that are commonly associated with abusive or harmful language. In the implementation phase, this trained model is used to evaluate new messages provided by users. The model analyzes the input features and produces a prediction indicating whether the message belongs to the cyberbullying category or the non-cyberbullying category. This prediction is generated quickly, allowing the system to provide near real-time feedback to the user.After the model generates a prediction, the system proceeds to the decision-making stage. In this stage, the output of the classification model is interpreted and an appropriate action is taken. If the model identifies the message as containing cyberbullying content, the system blocks the message and prevents it from being delivered. This mechanism helps discourage users from sending abusive messages and promotes responsible online communication. On the other hand, if the message is classified as non-cyberbullying, the system allows the message to be delivered normally.

The implementation also ensures that the system operates efficiently and responds quickly to user requests. Since the model is loaded during the startup of the application, the time required to process each message is significantly reduced. This allows the system to handle multiple user inputs smoothly without noticeable delays. The modular structure of the implementation further improves maintainability and scalability. Each component of the system, such as the user interface, preprocessing module, inference engine, and decision module, operates independently but works together to achieve the overall objective of cyberbullying detection.

Another important aspect of the implementation is the integration between the machine learning components and the web interface. The Streamlit framework simplifies this integration by allowing Python-based models to be directly connected with interactive web elements. This enables the system to display predictions instantly on the interface without requiring complex backend configurations. The implementation demonstrates how modern web technologies and artificial intelligence techniques can be combined to create intelligent applications that address real-world problems such as cyberbullying.

Overall, the implementation of the cyberbullying detection system successfully demonstrates the practical application of machine learning for online safety. By integrating natural language processing techniques, a trained classification model, and a user-friendly interface, the system provides an effective solution for detecting and preventing harmful communication. The implemented system can serve as a foundation for future improvements, such as incorporating larger datasets, using advanced deep learning models, or integrating the system with social media platforms and messaging applications. These future enhancements could further increase the accuracy and scalability of the system, making it more effective in combating cyberbullying in digital communication environments.

4.1 Pseudocode

Pseudocode is a simplified and informal way of representing the logic of a program or system using plain language and structured steps. It helps in explaining how a system works without using the exact syntax of any programming language. In software development, pseudocode is commonly used during the design and implementation phases to clearly describe the flow of operations in a system. It allows developers and readers to easily understand the sequence of steps involved in processing data and making decisions within the application.

In this project, pseudocode is used to represent the working procedure of the cyberbullying detection system. The pseudocode outlines the steps involved in receiving a user message, preprocessing the text, generating embeddings, and predicting whether the message contains cyberbullying content or not. It also explains how the system makes decisions based on the prediction result, such as blocking harmful messages or allowing safe messages to pass. By presenting the logic of the system in a structured format, the pseudocode helps in understanding the overall functionality of the implemented cyberbullying detection system.

The pseudocode describes the overall workflow of the cyberbullying detection system that utilizes both machine learning and transformer-based models to identify harmful text messages. The process begins by loading a labeled dataset containing messages that are categorized as cyberbullying or non-cyberbullying. Before training the models, the textual data undergoes several preprocessing steps to improve its quality and consistency. These preprocessing steps include converting all text into lowercase to maintain uniformity, removing punctuation marks and special characters that do not contribute meaningful information, eliminating stopwords that have little significance in classification tasks, and tokenizing the text into individual words. After completing these steps, the cleaned dataset is divided into training and testing sets so that the models can learn patterns from the training data and later be evaluated using unseen testing data.

Algorithm 4.1 CrossTongueGuard System using ML and Transformer Models

```
Input: Labeled text dataset, User message
Output: Cyberbullying or Non-Cyberbullying prediction
Load labeled cyberbullying dataset
Perform text preprocessing
Convert text to lowercase
Remove punctuation and special characters
Remove stopwords
Tokenize text data
Split dataset into training and testing sets
Machine Learning Model Training
Convert text into numerical feature vectors using TF-IDF
Select machine learning classifier
Train model using training dataset
Evaluate model performance on testing dataset
Save trained ML model
Transformer Model Training
Load pretrained transformer model
Tokenize dataset using transformer tokenizer
Convert tokens into embeddings
Fine-tune transformer model on training dataset
Evaluate transformer model accuracy
Save trained transformer model
Prediction Phase
Accept input message from user interface
Apply preprocessing to the input message
Convert message into feature vector for ML model
Tokenize message for transformer model
Generate prediction using ML model
Generate prediction using transformer model
Combine model outputs or select best model prediction
if prediction == Cyberbullying then
    Display warning message
    Block or flag harmful content
else
    Display message as safe
end if
Display final result on user interface
```

Following preprocessing, the system performs model training using two different approaches: traditional machine learning techniques and transformer-based deep learning models. For the machine learning approach, the textual data is converted into numerical form using the TF-IDF (Term Frequency–Inverse Document Frequency) technique, which measures the importance of words in the dataset. A suitable machine learning classifier is then selected and trained using the training dataset, after which its performance is evaluated on the testing dataset. At the same time, a pretrained transformer model is loaded and fine-tuned using the same dataset to capture contextual relationships between words within sentences. The dataset is tokenized using the transformer tokenizer and converted into embeddings that help the model understand the semantic meaning of the text. Once training and evaluation are completed, both models are saved for use during the prediction phase.

In the prediction phase, the system accepts a new message entered by the user through the application interface. The input message undergoes the same preprocessing steps applied during training to ensure consistency between training and prediction stages. The processed message is then transformed into a feature vector for the machine learning model and tokenized for the transformer model. Both models analyze the message and generate predictions regarding whether the content contains cyberbullying language. The system then combines the outputs of both models or selects the best-performing model to produce the final classification. If the message is identified as cyberbullying, the system displays a warning and flags or blocks the harmful content. Otherwise, the message is classified as safe, and the final result is displayed to the user through the interface

4.1.1 Pseudocode for Machine Learning Algorithms

The above pseudocode describes the procedure used for training and predicting cyberbullying messages using machine learning techniques. The process begins by loading the preprocessed dataset and converting the textual data into numerical feature vectors using the TF-IDF method. This method calculates the importance of each word in a document relative to the entire dataset, allowing the model to focus on words that are more relevant to classification. After feature extraction, the dataset is divided into training and testing sets so that the model can learn from the training data and be evaluated on unseen data.

During the training phase, the selected machine learning classifier learns the relationship between input features and their corresponding labels by adjusting its internal parameters. The model repeatedly predicts outputs for training samples, calculates the error, and updates its weights to minimize the classification loss. After training, the model is evaluated using the testing dataset to measure its accuracy and performance. In the prediction phase, a new user message is processed and converted into the same TF-IDF feature representation before being passed to the trained classifier.

Algorithm 4.2 Machine Learning Based Cyberbullying Detection

Input: Preprocessed text dataset
Output: Trained machine learning classifier
Load preprocessed cyberbullying dataset
Convert text documents into TF-IDF feature vectors
Compute Term Frequency:
$TF(t,d) = \frac{f_{t,d}}{\sum_k f_{k,d}}$
Compute Inverse Document Frequency:
$IDF(t) = \log\left(\frac{N}{n_t}\right)$
Compute TF-IDF weight:
$TFIDF(t,d) = TF(t,d) \times IDF(t)$
Split dataset into training set and testing set
Select machine learning classifier
for each training sample **do**
    Compute prediction using classification function
    $y = f(XW + b)$
    Calculate classification error
    $E = y_{true} - y_{pred}$
    Update model weights
    $W = W - \eta \nabla E$
end for
Evaluate model performance on testing dataset
Save trained model
Prediction Phase
Accept user input message
Apply preprocessing
Convert input text into TF-IDF feature vector
Predict class label using trained classifier
$\hat{y} = f(XW + b)$
if $\hat{y} =$ Cyberbullying **then**
    Display warning message
else
    Display message as safe
end if

4.1.2 Pseudocode for Transformer Models

The transformer-based cyberbullying detection algorithm uses a deep learning architecture to understand contextual relationships between words in a text message. The process begins by loading the labeled dataset and tokenizing the text using a transformer tokenizer. Each token is converted into a numerical embedding vector, which represents the semantic meaning of the words. Positional encoding is then added to the embeddings to preserve the order of words in the sentence, since transformer models do not inherently process sequences sequentially. The model then computes Query (Q), Key (K), and Value (V) matrices from the embeddings, which are used in the self-attention mechanism. The self-attention operation calculates attention scores using the formulas.

Algorithm 4.3 Transformer Model Based Cyberbullying Detection

Input: Preprocessed text dataset
Output: Trained transformer model
Load labeled cyberbullying dataset
Tokenize text using transformer tokenizer
Convert tokens into embeddings
Compute word embeddings:
$E = \text{Embedding(tokens)}$
Add positional encoding to embeddings
$Z = E + PE$
Split dataset into training and testing sets
Self-Attention Mechanism
Compute Query, Key, and Value matrices:
$Q = ZW_Q$
$K = ZW_K$
$V = ZW_V$
Compute attention scores:
$Attention(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$
Pass output through feed-forward neural network
$FFN(x) = \max(0, xW_1 + b_1)W_2 + b_2$
for each training epoch **do**
    Compute prediction probabilities using Softmax
    $P(y|x) = \text{softmax}(Wh + b)$
    Calculate loss using cross entropy
    $L = -\sum y \log(\hat{y})$
    Update model parameters using backpropagation
end for
Save trained transformer model
Prediction Phase
Accept user input message
Tokenize input text
Convert tokens into embeddings
Pass embeddings through transformer layers
Generate prediction using classification layer
if prediction = Cyberbullying **then**
    Display warning message
else
    Display message as safe
end if

After computing attention outputs, the representations are passed through a feed-forward network. During the training phase, the model generates prediction probabilities using a Softmax classification layer and calculates the loss using the cross-entropy loss function.The model parameters are then updated using backpropagation to minimize the loss and improve prediction accuracy. Once training is complete, the transformer model is saved and used in the prediction phase. When a user enters a new message, it is tokenized, converted into embeddings, and passed through the transformers to generate a classification result indicating whether the message contains cyberbullying or not.

4.2 Modules

The proposed CrossTongueGuard system is organized into several functional modules that work together to process textual data and accurately identify harmful or abusive content. Each module performs a specific task in the overall workflow, starting from data acquisition and preprocessing to feature extraction, model training, and final prediction. Dividing the system into modules improves the clarity, scalability, and efficiency of the implementation. It also allows different components of the system to be developed and evaluated independently while contributing to the overall objective of detecting cyberbullying in online text.

These modules collectively ensure that the input data is properly prepared, transformed into meaningful numerical representations, and analyzed using both machine learning algorithms and transformer-based deep learning models. The modular design enables the system to handle large datasets, improve classification accuracy, and provide reliable predictions. The following subsections describe each module of the system in detail, including their functionality and role in the cyberbullying detection process.

4.2.1 Data Collection

The Data Collection module is responsible for gathering cyberbullying-related datasets from various online sources such as social media platforms and publicly available datasets. This data contains labeled examples of bullying and non-bullying text that are used for training and evaluation. The youtube parsed dataset is downloaded from kaggle platform. And the text is extracted from the dataset and created new csv file.

4.2.2 Data Preprocessing

The Text Preprocessing Module prepares the collected data for analysis. In this stage, raw text messages are cleaned and normalized through several steps such as removing punctuation, converting text to lowercase, eliminating stop words, and applying tokenization. These preprocessing steps reduce noise in the dataset and make the text suitable for feature extraction and model training.

4.2.3 Feature Extraction

The Feature Extraction Module demonstrates textual data is transformed into numerical representations that machine learning models can understand. Techniques such as TF-IDF (Term Frequency–Inverse Document Frequency) are used to convert words into weighted feature vectors that represent their importance in the dataset. This transformation allows the algorithms to process textual information mathematically.

4.2.4 Machine Learning Module

The Machine Learning Classification Module is used to train traditional machine learning models to detect cyberbullying content. Algorithms such as Logistic Regression, Support Vector Machine, or Naïve Bayes can be used to learn patterns from the extracted features. The model analyzes the feature vectors and predicts whether a given message belongs to the cyberbullying class or the non-cyberbullying class.

4.2.5 Transformer Module

The Transformer-Based Deep Learning Module utilizes advanced transformer models to capture contextual relationships between words in a sentence. Unlike traditional models, transformer architectures use attention mechanisms to focus on important words and understand the meaning of the entire sentence. This module improves classification performance by analyzing deeper semantic relationships in the text.

4.2.6 Prediction and Alert Module

The Prediction and Alert Module is responsible for real-time detection of cyberbullying messages. When a user inputs a message, the system preprocesses the text, converts it into features, and passes it through the trained model. Based on the prediction result, the system identifies whether the message contains cyberbullying content and displays a warning or allows the message if it is safe.

Chapter 5

RESULTS AND DISCUSSION

The results obtained from the proposed CrossTongueGuard system are analyzed and discussed in this section. Various visualization techniques and performance evaluation metrics are used to understand the characteristics of the dataset and the effectiveness of the machine learning and transformer-based models. Initially, exploratory data analysis is performed to study the distribution, relationships, and statistical behavior of the dataset features. Visualization techniques such as class distribution, boxplots, probability density function (PDF) curves, and correlation matrices help in understanding the dataset structure and identifying patterns or imbalances. After preprocessing and feature extraction, multiple machine learning and transformer models are trained and evaluated.

The performance of these models is assessed using evaluation metrics such as accuracy, precision, recall, and F1-score. Graphical representations including confusion matrices, ROC curves, and comparison graphs are used to visually interpret the classification performance. In addition, the developed user interface is demonstrated to show how the system allows users to input text and obtain cyberbullying detection results in real time.This practical implementation demonstrates the real-world applicability of the system in monitoring and moderating harmful content on online communication platforms.

The results obtained from the experiments indicate that the combination of machine learning techniques and transformer-based models can effectively detect cyberbullying language and contribute to creating safer digital communication environments. The functionalities of the Logistic Regression, Naive Bayes, and Support Vector Machine classifiers were tested using TF-IDF features.

By taking advantage of the lexical patterns in the text, these models were capable of distinguishing cyberbullying from non, cyberbullying content at an effective baseline level of the traditional methods, Support Vector Machine showed relatively stronger discrimination ability because it is able to find the best decision boundaries in a high, dimensional feature space.

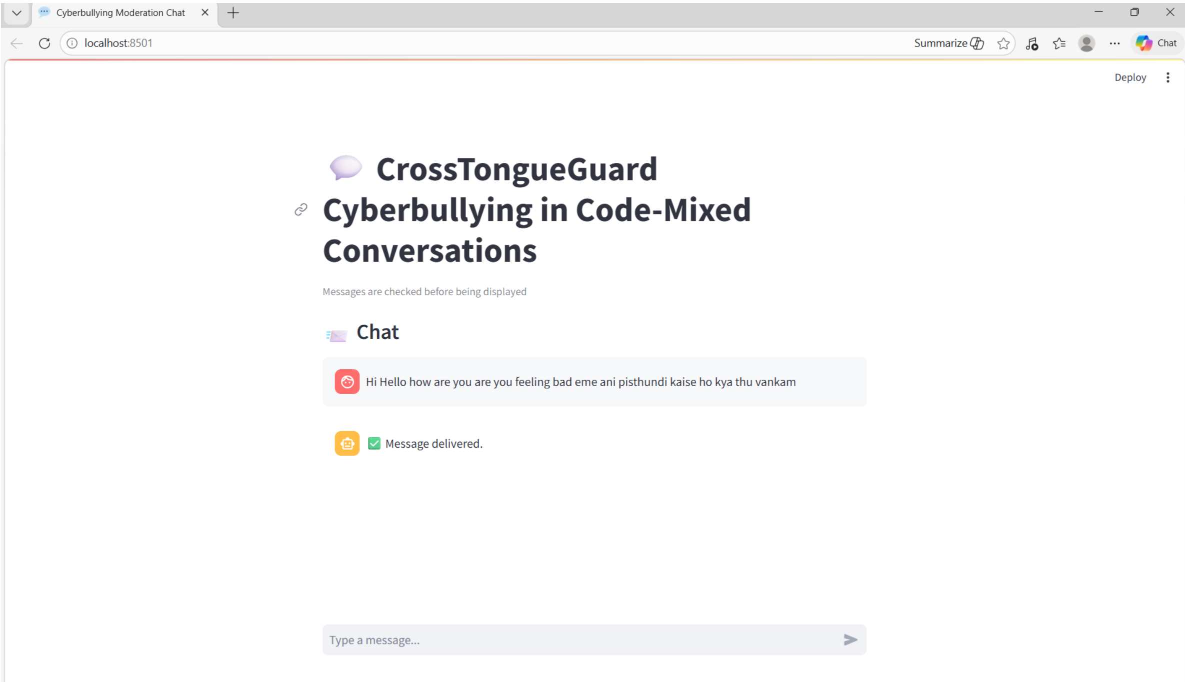

Figure 5.1: Non-Cyberbulling Output

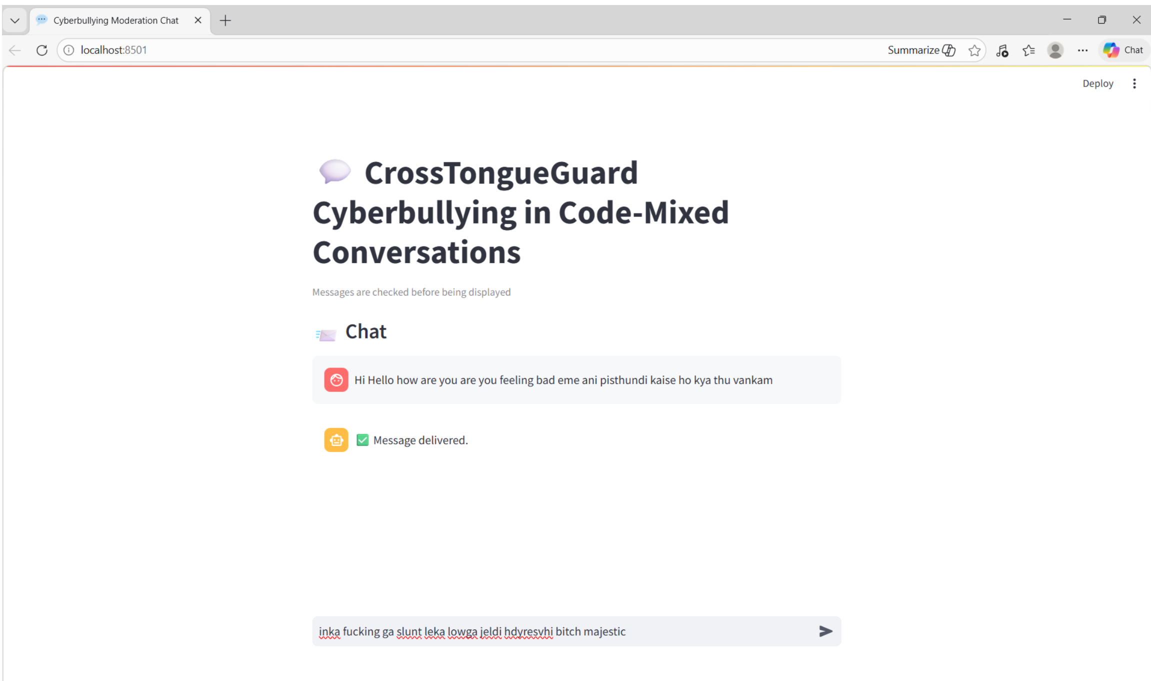

Figure 5.2: Enter any Message

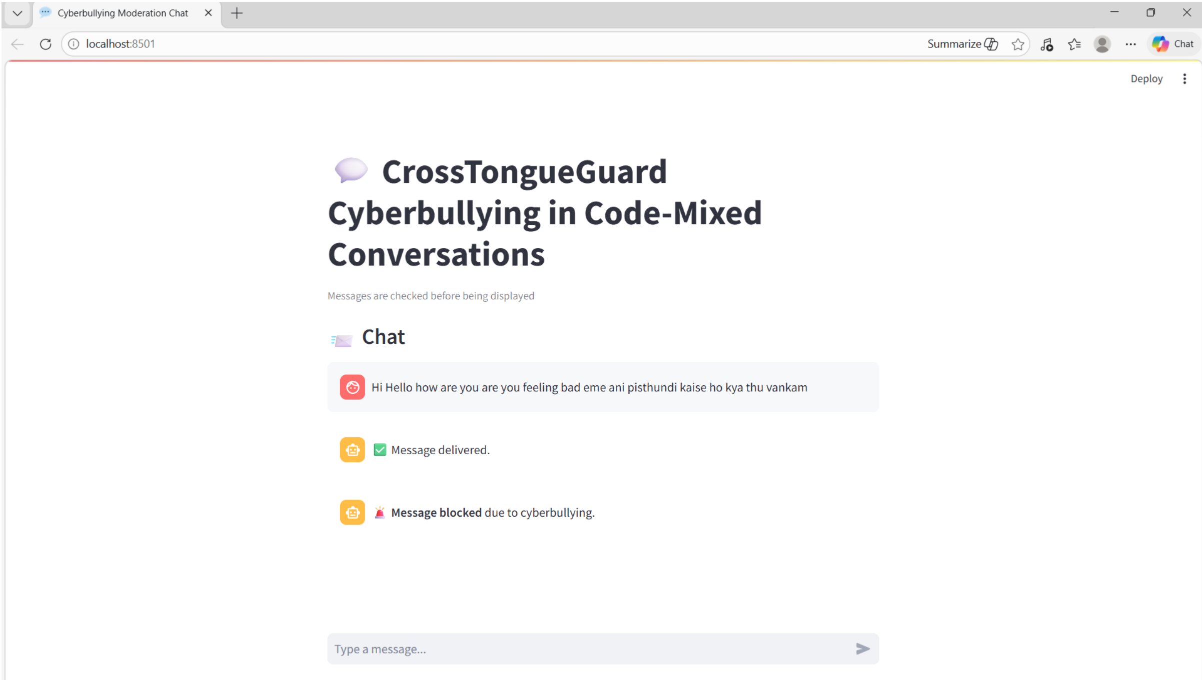

Figure 5.3: Cyberbulling Output

The confusion matrix is a tool that provides a detailed view of the classification results, such as true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN). This kind of analysis helps in understanding the classification patterns and behaviour of the model.

As mentioned in figure 5.4, Logistic Regression correctly classified 291 instances (TN = 155, TP = 136) but resulted in 45 false negatives, meaning that many instances of cyberbullying were classified as non-cyberbullying. Although the model performs decently, the slightly higher false negatives might affect its applicability in real-world moderation systems. The SVM classifier enhanced the separation of classes, obtaining TN = 159 and TP = 144 with fewer false positives (22) and false negatives (37). Naïve Bayes performed well in detection capability for cyberbullying instances (TP = 171, FN = 10), resulting in high recall. However, it resulted in 121 false positives, significantly affecting precision. This shows that the model over-classified the content as cyberbullying, which is likely due to the assumption of independence in the model.

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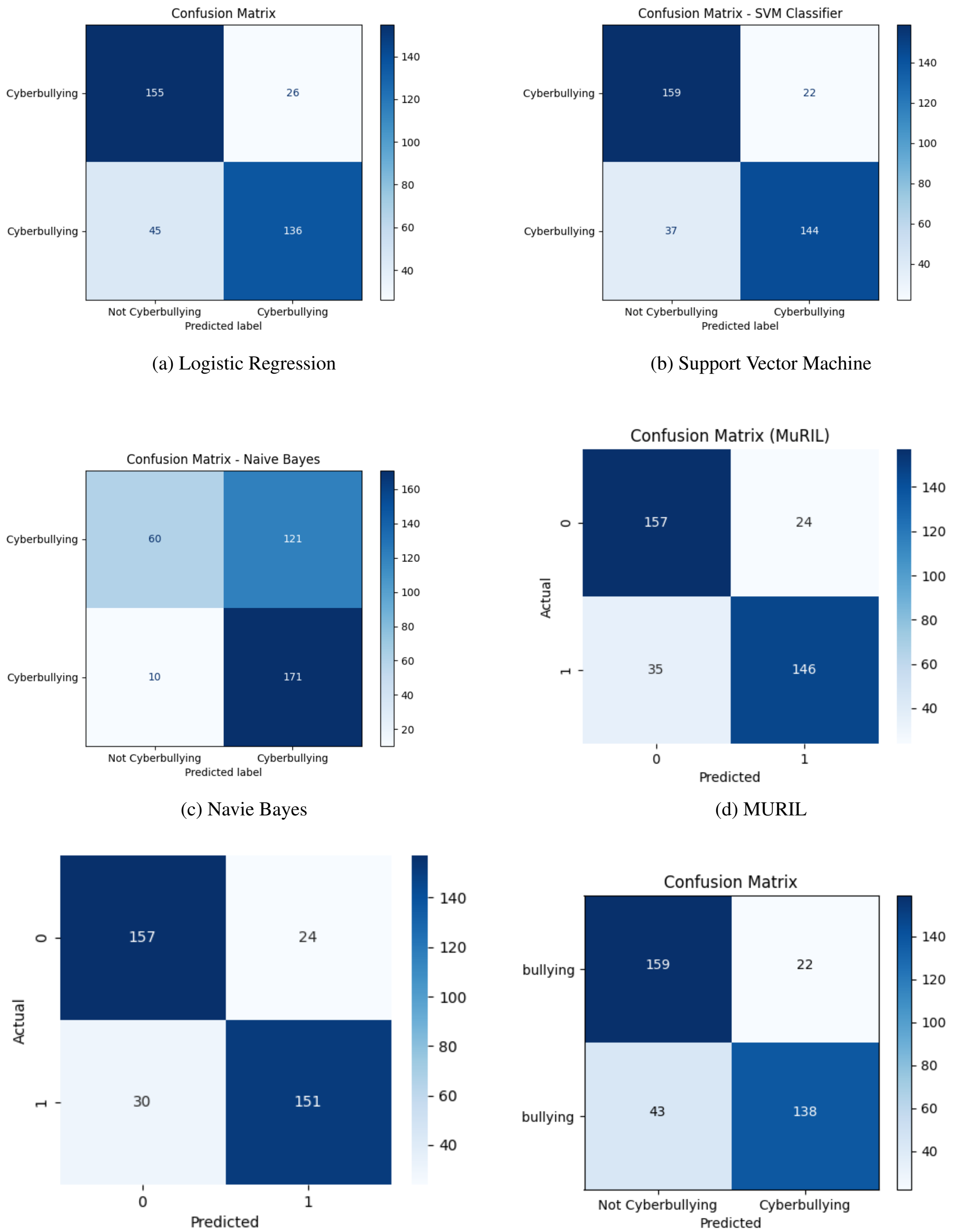


(a) Logistic Regression

(b) Support Vector Machine

(c) Navie Bayes

(d) MURIL

(e) XLM-RoBERTa

(f) mDeBERTa

Figure 5.4: Confusion Matrices of Various Algorithms

MuRIL performed well in balanced classification results (TN = 157, TP = 146) with moderate false positives (24) and false negatives (35). The multilingual contextual embeddings used in the model help in better semantic understanding than traditional machine learning models. XLM-RoBERTa performed one of the best (TN = 157, TP = 151) with only 30 false negatives, meaning that the model has better contextual understanding and a strong cyberbullying detection capability. mDeBERTa performed competitively (TN = 159, TP = 138), although it resulted in slightly higher false negatives (43) than XLM-RoBERTa and MuRIL.

Quantitative performance was assessed by metrics like Accuracy, Precision, Recall, and F1, score. Precision and Recall played a crucial role in this study because cyberbullying detection necessitates not only a reduction in false negatives but also a retention of the system's resistance against false positives. The F1, score gave a balanced view of the model's performance.

Table 5.1: Algorithm Performance Comparison

Algorithm	**Accuracy**	**Precision**	**Recall**	**F1-Score**
Logistic Regression	0.80	0.83	0.75	0.79
SVM	0.83	0.86	0.79	0.82
Naive Bayes	0.63	0.58	0.74	0.72
MURIL	0.83	0.85	0.80	0.83
XLM-RoBERTa	0.85	0.86	0.83	0.84
mDeBERTa	0.82	0.86	0.76	0.80

XLM-RoBERTa had the highest overall accuracy of 85% and F1-score of 0.84. In the traditional machine learning models, SVM had the highest accuracy of 84% and F1-score of 0.83. Logistic Regression had a moderate F1-score of 0.79. Naïve Bayes had the highest recall of 0.94, which shows high sensitivity to the cyberbullying content. However, its precision of 0.59 was much lower, which shows a high false positive rate and lower reliability for use in real-world systems.

The performance comparison of various algorithms (Figure 5.2,5.3,5.4,5.5) further shows that transformer models have a clear edge over traditional models in most of the performance metrics such as accuracy,precision,recall and f1-score.Therefore, XLM-RoBERTa got the best performance among all the models in terms of accuracy,precision,recall and f1-score.

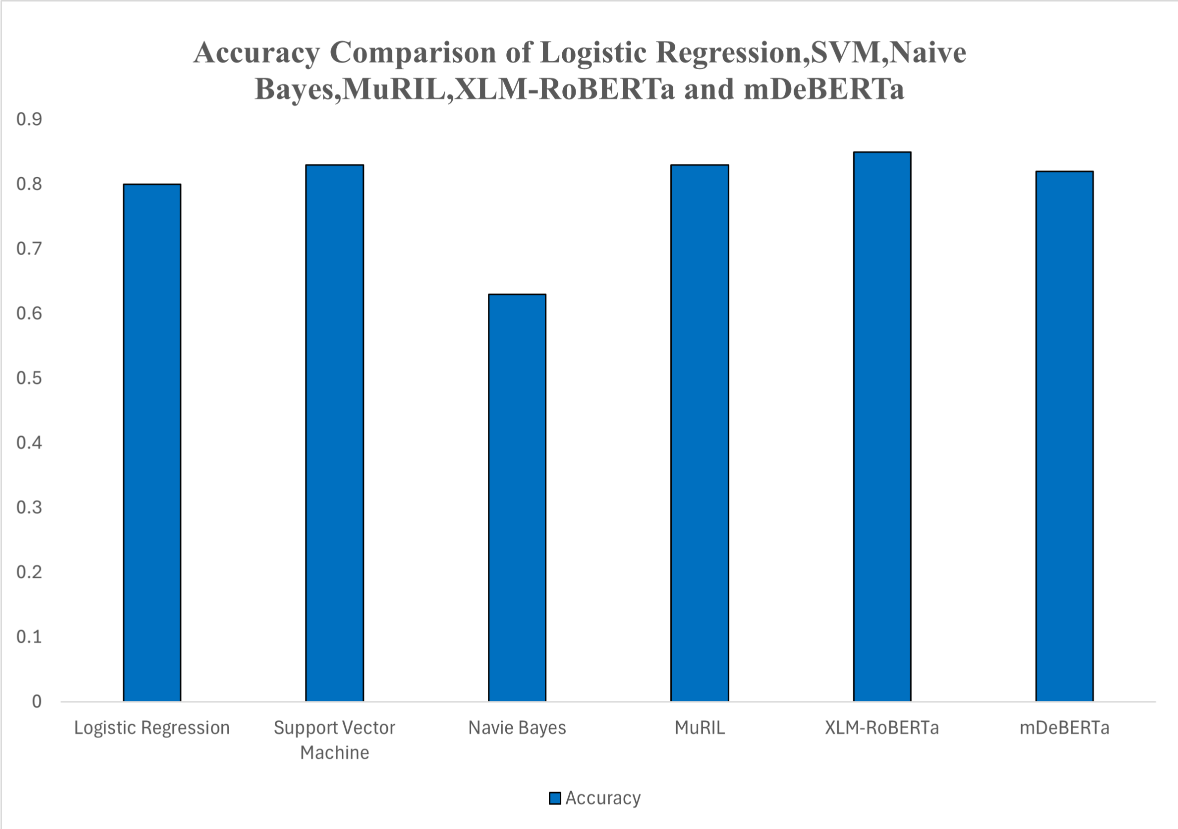


Figure 5.5: Accuracy Comparison Graph

The Accuracy comparison graph figure 5.5 shows that transformer-based models outperform traditional machine learning models. Among all models, XLM-RoBERTa achieves the highest accuracy, indicating its superior ability to correctly classify both cyberbullying and non-cyberbullying instances. Support Vector Machine (SVM) performs best among the traditional models, followed by Logistic Regression, while Naïve Bayes shows comparatively lower accuracy due to its higher false positive rate.

The Precision comparison graph figure 5.6 highlights the model's ability to correctly identify only relevant cyberbullying instances. XLM-RoBERTa and mDeBERTa achieve the highest precision values, indicating fewer false positive predictions. Naïve Bayes exhibits significantly lower precision, which suggests that it tends to incorrectly classify non-cyberbullying content as cyberbullying.From the graph, it can be observed that transformer-based models such as XLM-RoBERTa, MuRIL, and mDeBERTa achieve consistently high precision scores. Among them, XLM-RoBERTa demonstrates the highest precision, indicating its strong capability to accurately distinguish between harmful and non-harmful content. This high precision is mainly due to the model's ability to capture contextual and semantic relationships within the text, reducing misclassification.

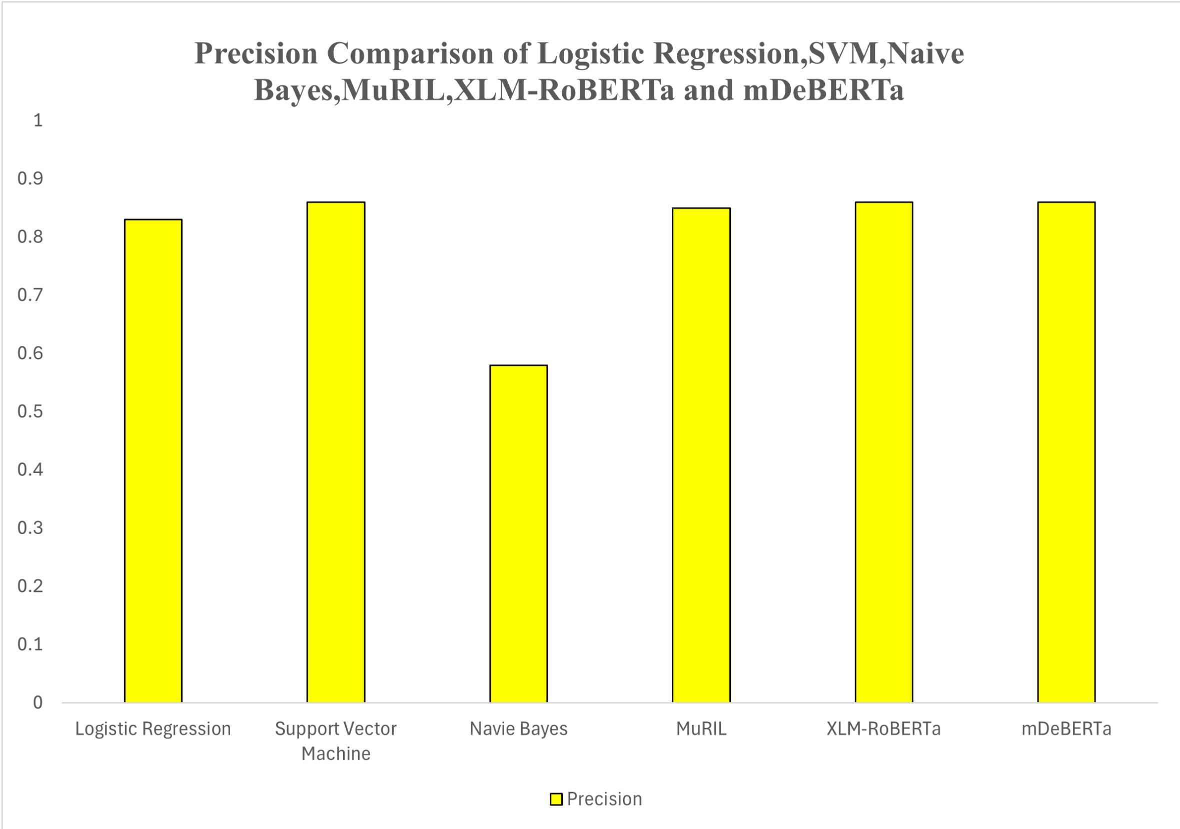


Figure 5.6: Precision Comparison Graph

The Recall comparison graph figure 5.7 measures the ability of the models to detect actual cyberbullying instances. XLM-RoBERTa again achieves the highest recall, demonstrating its strong capability in identifying harmful content. Naïve Bayes also shows relatively high recall, meaning it is sensitive to cyberbullying detection; however, this comes at the cost of lower precision.From the graph, it is evident that transformer-based models, particularly XLM-RoBERTa, achieve the highest recall among all the models. This demonstrates their strong capability to detect a large proportion of cyberbullying instances. The contextual understanding provided by transformer architectures allows them to identify subtle forms of abusive language, sarcasm, and implicit harmful intent that traditional models may fail to capture.MuRIL and mDeBERTa also show competitive recall values, indicating their effectiveness in handling multilingual and context-rich data.

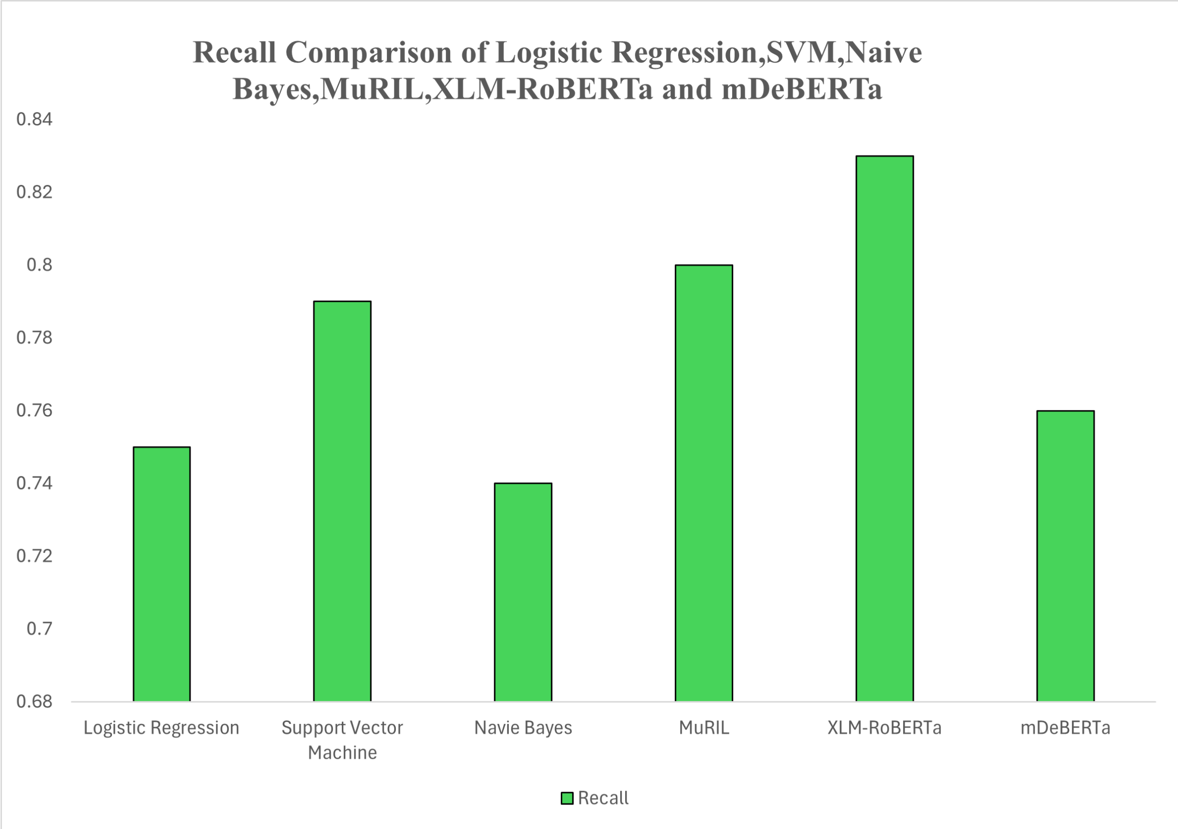


Figure 5.7: Recall Comparison Graph

The F1-Score comparison graph figure 5.8 provides a balanced evaluation by considering both precision and recall. XLM-RoBERTa achieves the highest F1-score, indicating a strong balance between detecting cyberbullying content and minimizing false predictions. Transformer models such as MuRIL and mDeBERTa also demonstrate competitive performance, while traditional models like Logistic Regression and SVM show moderate results.

From the graph, it is evident that transformer-based models outperform traditional machine learning models in terms of F1-score. Among all models, XLM-RoBERTa achieves the highest F1-score, indicating an optimal balance between precision and recall. This means that the model is both highly accurate in detecting cyberbullying content and efficient in avoiding false predictions.

MuRIL and mDeBERTa also demonstrate strong F1-score performance, showing that transformer-based architectures are generally more effective in handling complex language patterns and contextual information. These models are capable of capturing semantic relationships between words, which significantly improves their overall classification performance.

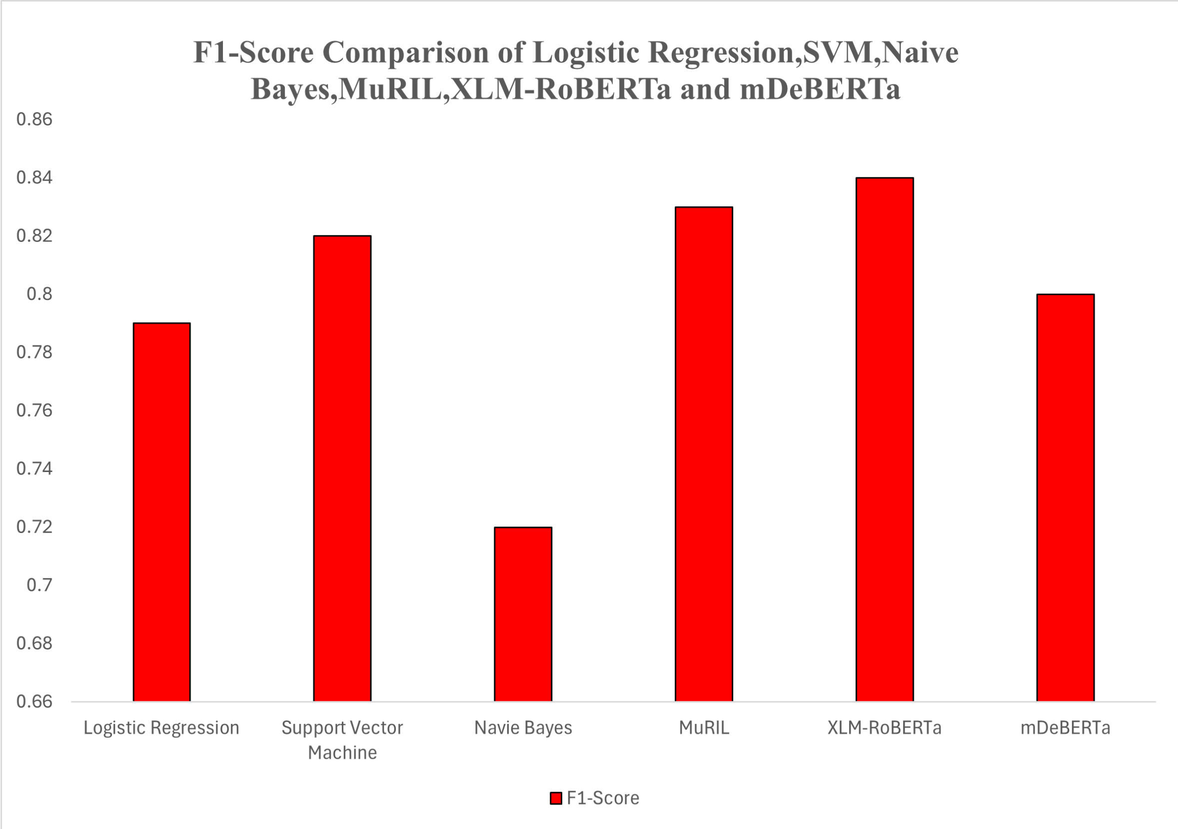
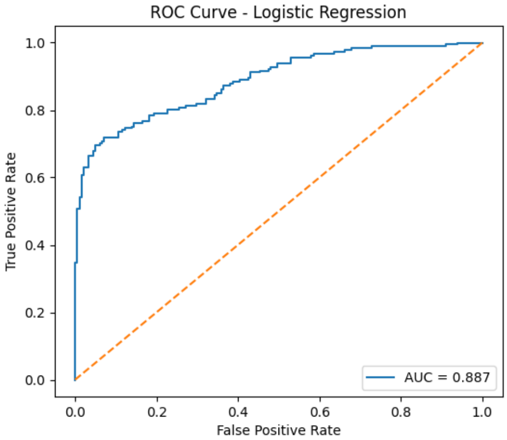


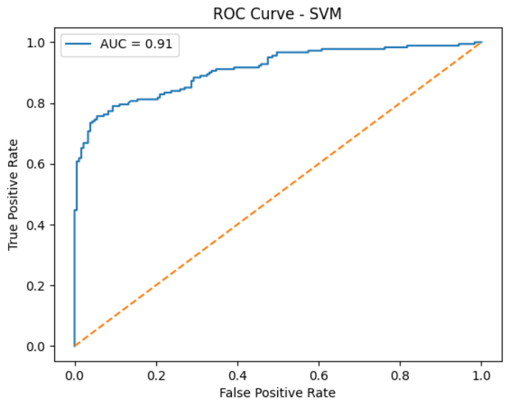
Figure 5.8: F1-Score Comparison Graph

The Receiver Operating Characteristic (ROC) curve is used to evaluate the classification performance of the cyberbullying detection models. The ROC curve represents the relationship between the True Positive Rate (TPR) and the False Positive Rate (FPR) at different classification thresholds. A model with a curve closer to the top-left corner indicates better classification performance. The Area Under the Curve (AUC) is used as a quantitative measure of the model's ability to distinguish between cyberbullying and non-cyberbullying instances. Higher AUC values indicate stronger discriminative capability of the model.

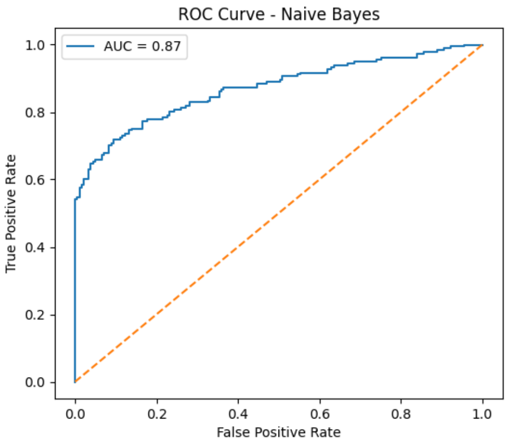
As from the figure 5.9, the transformer-based models excel in distinguishing classes better than other machine learning techniques. XLM-RoBERTa dominates the chart by achieving the highest Area Under the Curve (AUC). This confirms that XLM-RoBERTa discriminates best among classes for different classification thresholds. MuRIL and mDeBERTa also achieve high AUC values, which validates the contextual embedding technique's effectiveness.



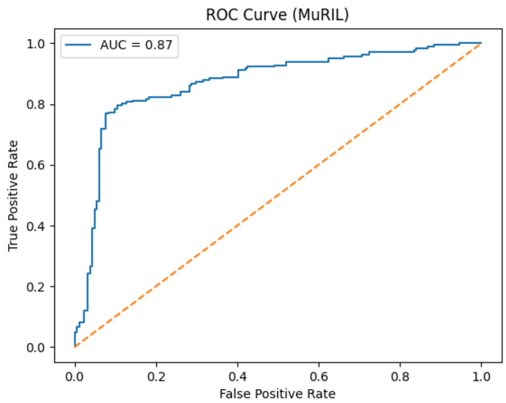
(a) Logistic Regression



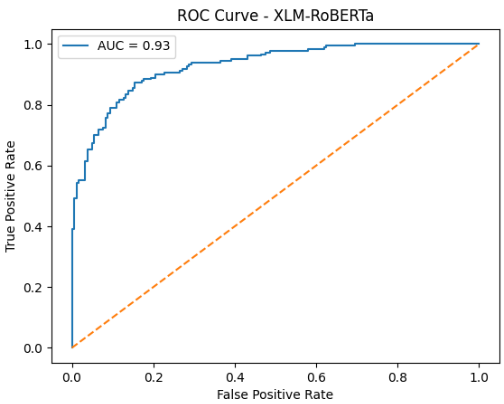
(b) Support Vector Machine



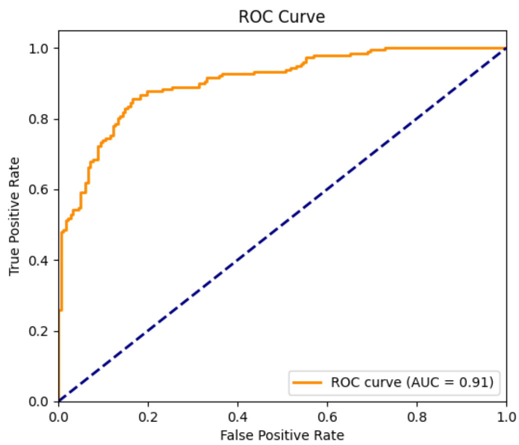
(c) Navie Bayes



(d) MURIL



(e) XLM-RoBERTa



(f) mDeBERTa

Figure 5.9: ROC Graphs of Various Algorithms

The best classical models for classification are SVM, followed by Logistic Regression and Naive Bayes. Naive Bayes achieves a high recall rate but shows instability in the ROC curve, implying a higher rate of false positives. The higher AUC for transformer-based models validates their capability to comprehend contextual nuances such as sarcasm and aggressiveness inherent in cyberbullying.

The results show that contextual transformer models outperform traditional machine learning techniques by a good margin when it comes to detecting cyberbullying. Among them, XLM-RoBERTa emerges as the most balanced and reliable option, which not only offers high accuracy, a high F1-score, and a high AUC, but also manages to keep the false negative rate low. On the other hand, SVM emerges as a reliable baseline option, while Naïve Bayes seems to over-predict cyberbullying, making it less viable in real-world scenarios.

Overall, this proves that adding deep contextual representations can enhance the ability of the system to robustly and semantically detect harmful content.

The results obtained from the experimental evaluation demonstrate the effectiveness of both traditional machine learning models and transformer-based models in detecting cyberbullying content. From the analysis of performance metrics, it can be observed that transformer models such as XLM-RoBERTa outperform traditional algorithms in terms of accuracy and F1-score. This improvement can be attributed to the ability of transformer models to capture contextual relationships and semantic meanings within text data.

In contrast, traditional machine learning algorithms rely mainly on feature extraction techniques such as TF-IDF, which may not fully capture the contextual dependencies in natural language. Among the traditional models, Support Vector Machine (SVM) showed competitive performance, indicating its suitability for text classification tasks. Naïve Bayes demonstrated high recall, which suggests that it is highly sensitive in detecting cyberbullying instances; however, its lower precision indicates the presence of more false positive predictions. The graphical analysis through ROC curves, confusion matrices, and performance comparison graphs further validates the effectiveness of the proposed system.

Chapter 6

CONCLUSION AND FUTURE SCOPE

This project presents a comprehensive comparative analysis of traditional machine learning classifier models and transformer-based models for cyberbullying detection. The traditional classifier models considered in this work are Logistic Regression, SVM, and Naïve Bayes. On the other hand, multilingual transformer-based models, such as MuRIL, XLM-RoBERTa, and mDeBERTa, are considered. The performance of each classifier was tested using accuracy, precision, recall, and F1-measure, confusion matrix, and ROC-AUC curve. Among all the models considered in this work, XLM-RoBERTa was found to perform the best. It had the highest accuracy, F1-measure, and Area Under the Curve (AUC). Its low number of false negatives also makes it more reliable in identifying harmful content. Although SVM performed relatively well, Naïve Bayes performed well in terms of high recall value, its high number of false positives makes it less applicable.

Despite the high performance of transformer-based models, there are many opportunities and directions for further research. Another direction is using hybrid models, which can be made by combining transformer embedding and other state-of-the-art models like attention. Another direction is using different optimization techniques like knowledge distillation, which can be used to reduce the complexity of the models. Another direction is using explainable AI, which can be used to improve transparency and interpretability. This work presents a comprehensive comparative analysis of traditional classifier models and transformer-based models. It demonstrates the effectiveness of transformer-based models in identifying cyberbullying.

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APPENDIX

ACCEPTANCE LETTER

9th International Conference on
Inventive Computation Technologies
ICICT 2026
15-17, April 2026

icict.conf@gmail.com
https://icicts.com/2026/

Acceptance Letter

Paper ID	Title	Author(s)
ICICT-1343	CROSSTONGUEGUARD: DETECTING CYBERBULLYING IN CODE-MIXED CONVERSATIONS	Kadaganchi Sudhakar, Nakka Upender Goud, Bhogolu Denesh Kumar, Karamkonda Srinadh, Pathlavath Bhaskar

Dear Author(s),

We are pleased to inform you that your paper has been accepted for presentation and publication at the 9th International Conference on Inventive Computation Technologies (ICICT 2026), to be held in Lalitpur, Nepal, during April 15–17, 2026. Your manuscript was reviewed by subject experts, and based on the reviewers' recommendations, it has been selected for inclusion in the conference proceedings. The accepted papers will be submitted for publication in IEEE Xplore, subject to IEEE compliance requirements. Authors are requested to complete the final submission and registration process within the prescribed deadlines.

Conference Publication Details:
ICICT ISBN: IEEE XPLORE COMPLIANT ISBN: 979-8-3315-6917-4
IEEE DVD ISBN: 979-8-3315-6916-7
IEEE Conference ID: #68280

We congratulate you on your acceptance and look forward to your participation in ICICT 2026.

Dr. Subarna Shakya
Conference Chair
ICICT 2026